

Module 2: The Federal Reserve

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Lesson 2-0 Module 2 Information

[Lesson 2-0.1: Overview](#)



Hello and welcome. I hope you're doing well from wherever it is you're joining us.

Module overview

- Federal Reserve System
- History, organization, purpose
- Monetary policy tools

In this module, we study the Federal Reserve System, its functions, and its monetary policy tools. We start with the history of the Federal Reserve. Why do we have a central

bank? How is it organized? What purpose does it serve? We will take a careful look at two goals of monetary policy: maximum employment and stable prices. This is often called the Dual Mandate of the Federal Reserve. For all our discussion for the remainder of this course should somehow relate back to the Dual Mandate. We will then examine the institutional features of the Federal Reserve and see how its monetary policy decision-making process works. To be clear, this course examines the Federal Reserve System of the United States as a case study. There are, of course, many other central banks around the world. Some central banks have a different mandate or a slightly different set of key functions. However, all of them conduct monetary policy and provide a payment system. In this course, we focus mostly on how central banks conduct monetary policy. Over time, there has been much disagreement about the best ways to conduct monetary policy. Some central banks explicitly target the money supply rather than interest rates. The Federal Reserve used to do this.

Traditional monetary policy

- Price of money = interest rate
- Target money supply?

Why do we care about money supply? Like any other good, the price of money is determined by demand for money and supply of money in the economy. What's perhaps unusual is that the price of money is the interest rate. Rather than targeting interest rates directly, it might make sense to control the money supply instead. But that begs the question, how can we measure the money supply? We will therefore introduce the concept of money supply, its measurement, and how it can be affected by monetary policy. We then examine the balance sheet of the Federal Reserve and how it has evolved over time. We will provide some insights about how the conduct of monetary policy has changed money supply.

Traditional monetary policy

- Price of money = interest rate
- Target money supply?
- Target interest rates directly?
 - Equilibrium in Fed Funds market
 - Open market operations
 - Discount window loans

To understand how exactly money supply and interest rates are related, we will focus on traditional monetary policy. We will discuss how open market operations directly affect the money supply in the federal funds market. We will see how these open market operations determine the equilibrium interest rate in this market. The Federal funds rate. The Federal funds rate is the main target interest rate of monetary policy in the US. A complimentary traditional monetary policy tool is the discount window. This policy tool provides short-term liquidity in form of short-term loans to banks. This tool is mainly designed for emergency uses so the interest rate is set higher than the Federal funds rate. But the discount window loans remain an alternative to Federal funds. The interest rate on the discount window loans, therefore, enforces the upper limit of the Federal funds rate target range set by the Federal Reserve.

New approaches

- Forward guidance
- Large-scale asset purchases
- Equilibrium in repo market

The last part of this module looks at new monetary policy tools that have emerged after the 2008 global financial crisis. These are often called unconventional monetary policy tools. The first is forward guidance. Forward guidance gives market participants a clear path for interest rates, or more generally, monetary policy actions in the future and can be used to target long-term interest rates. The second new tool are large-scale asset purchases. When the short-term interest rate is already zero, central banks can further increase the money supply by buying large amounts of government-backed debt such as Treasury securities and agency mortgage-backed securities. This approach allows the Central Bank to bring down long-term interest rates directly. The third new tool is the overnight reverse repurchase program or ON RRP. This tool is used to set interest rates in the repo market. The repo market is like the federal funds market and overnight lending market. Financial institutions could switch between these markets. They are substitutes. By managing the equilibrium in the repo market, the Federal Reserve ensures that all short-term interest rates are consistent with their monetary policy targets.

To conclude, this module provides an in-depth discussion of monetary policy tool. Many of these videos will require an understanding of the financial markets and instruments discussed in the first module. Specifically, the federal funds market, the repo market, Treasury securities, and the yield curve. You don't need to understand all the technical details of each monetary policy tool. However, I do want you to keep these two main high-level points in mind. First, the Federal Reserve targets multiple closely linked markets with its interest rate policy. Second, when the short-term interest rate is zero,

the central bank is not out of options. It can use forward guidance and large-scale asset purchases to target longer-term interest rates.

Lesson 2-1: The Federal Reserve

Lesson 2-1.1. History and Structure of the Federal Reserve



Learning Objectives

- What is a central bank?
- Purpose and functions of central banks
- Federal Reserve System:
The U.S. Central Bank

Hello and welcome to this lecture on the history and structure of the Federal Reserve System. In this class, we will introduce the concept of a central bank. Central banks exist all around the world. We will discuss the reasons why they were founded, what are their objectives and key functions. We will then focus on the Federal Reserve System, the Central Bank of the United States. We will discuss its foundation, we will examine the structure of the Federal Reserve system, and its decision-making bodies.

Case Study: Swedish Riksbank



Swedish Riksbank, n.d.
<https://www.riksbank.se/en-gb/about-the-riksbank/history/historical-timeline/1600-1699/first-banknotes-in-europe/>



Let's start by looking at the world's oldest central bank. This is the Swedish Riksbank, which is based in Stockholm, Sweden. It is a fun case study that helps us to understand why country's founded central banks. It tells us what they hope to achieve.

Case Study: Swedish Riksbank

Heavy metal coins as money

New idea: paper money issued by banks

Backed by coins held by bank

Redeemable on demand



We are in Sweden in 1661. At the time, money took the form of heavy coins made from precious metals. These coins could weigh as much as 45 pounds, not very convenient. In 1661, the Swedish king allowed private banks to print paper bank notes. This is what

we all know today as paper money. This is much more convenient for banks and citizens. Why were people willing to accept these banknotes? Well, because the bank promised to exchange its banknotes against coins deposited in the bank.



The bank notes were redeemable on demand. The banknotes were a huge success, but then the bank printed more and more banknotes. This led people to question whether the private bank would have enough coins deposited in its vaults to exchange all banknotes. These doubts led to an increasing number of people redeeming their banknotes for coins. This is an example of a bank run. However, the bank did not have enough coins to redeem its banknotes all at once. As a result, the bank collapsed.

Case Study: Swedish Riksbank

Problem: too much paper money

Solution:

Swedish Riksbank

Central bank controls
money supply



Despite this bank run, the idea of paper money was still popular. But it was understood that there was too much paper money in circulation relative to deposited coins. There was too much money relative to the amount of collateral held by the banking system. Hence, the Swedish Riksbank was founded in 1668 to control the money supply. The law required this new central bank to maintain the domestic coinage at its right and fair value. The law prohibited the Riksbank to print paper money at volumes that would cause inflation.

Reasons for Central Banks

1. Control money supply $\Rightarrow$ maintain price stability
2. Lender of last resort $\Rightarrow$ stop bank runs



The story of the foundation of the Swedish Riksbank showcases two important issues that modern central banks are designed to address. The first is ensuring price stability. Central banks are typically given control over the money supply. This control allows central banks to prevent too much paper money being printed. When there is too much money in the economy, this tends to cause price inflation. The second reason is that banks are susceptible to runs by fearful customers. To deal with bank runs, central banks act as a lender of last resort. Let me explain what this means. Banks borrow short term and lend long term. In our example, the Swedish bank borrowed metal coins in exchange for banknotes. It then used the coins in part to give out loans. At any point, there were fewer coins in the vault than banknotes outstanding. When everybody wanted their coins for their banknotes, the bank lacks the coins and could collapse. In modern banking, banks borrow cash from depositors and lend it out long term. They could also face a shortage of cash from time to time, even if they are making profitable loans to good borrowers.

Lender of Last Resort Function

1. Provides cash loan
2. Accepts illiquid assets as collateral



The lender of last resort function of central banks is designed to prevent the collapse of a healthy bank due to a shortage of cash. How does this work in practice? The Central Bank provides cash to the banks in form of short-term loans against collateral. This can happen during an emergency when the bank is stressed. Central banks typically accept many different types of collateral. Importantly, this can be illiquid long-term loans, that is, assets that would be hard to sell or that can only be sold with large discounts. For instance, today the Federal Reserve system accepts real estate and consumer loans as collateral for cash loans to banks.

First Bank of the United States (1791–1811)



Federal Reserve History. (2015). *Girard's Bank, late the Bank of the United States, in Third Street, Philadelphia* [Painting]. <https://www.federalreservehistory.org/essays/first-bank-of-the-us>

Let's now examine why the Federal Reserve system was founded. The Federal Reserve system is not the first US central bank, it is actually the third. There was the First Bank of America, from 1791 to 1811 and the Second Bank of America, from 1816 to 1836. However, these two earlier central banks were very different from what we today would consider to be a central bank. For instance, they had no control over the money supply.

No U.S. Central Bank (1837-1913)

Banking crises throughout this period

Panic of 1907

The United States had no central bank between 1837 and 1913. Money was issued

privately, just like in our Swedish example. In this period, there were several banking crises, including a panic in 1907. During the 1907 panic, the stock market lost 50% of its value. Banks nationwide saw customers lining up to withdraw their deposits.

No U.S. Central Bank (1837-1913)



Federal Reserve History. (2015). *Crowd on Wall Street during the Panic of 1907* [Photograph].
<https://www.federalreservehistory.org/essays/panic-of-1907>



The panic of 1907 illustrated the need for a lender of last resort. But it took some time and much political debate until the Federal Reserve system was officially established.

Federal Reserve System

Established in 1913



Kurtz, W. G. (1923). *President Wilson signing the Federal Reserve Act* [Painting]. Federal Reserve History. <https://www.federalreservehistory.org/essays/federal-reserve-act-signed>

On December 24th, 1913, President Woodrow Wilson signed the Federal Reserve Act.

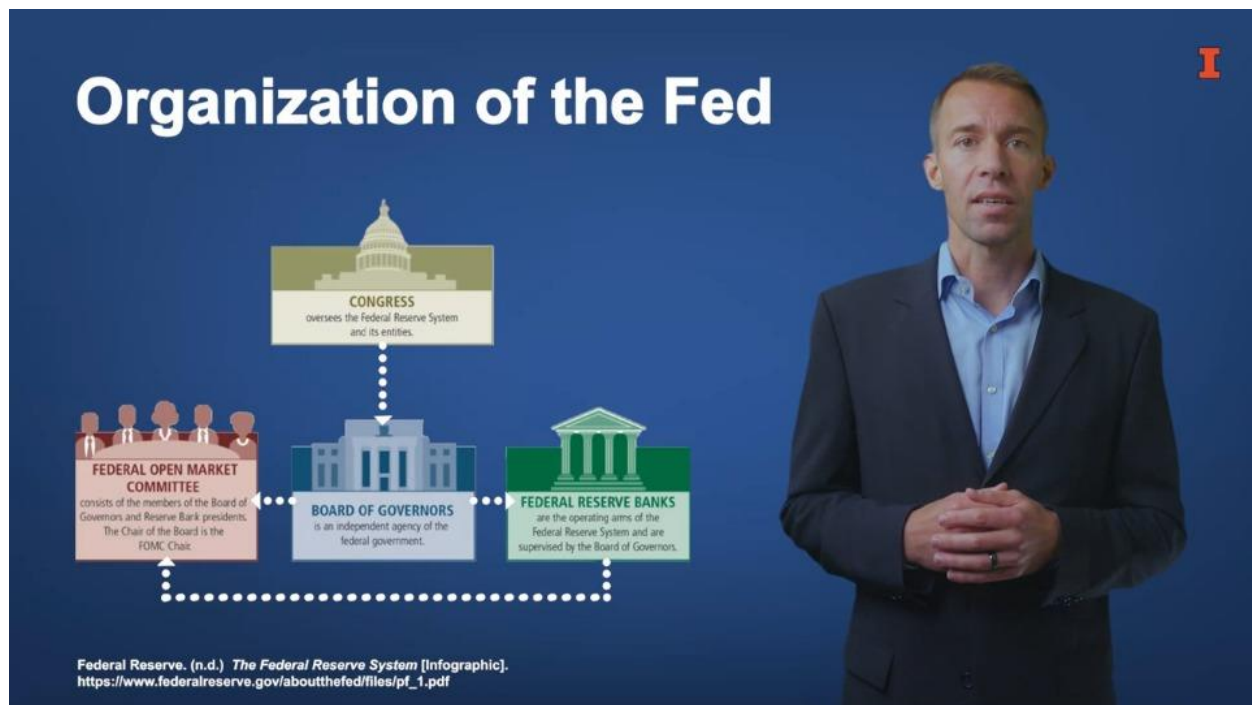
Federal Reserve System

Established in 1913

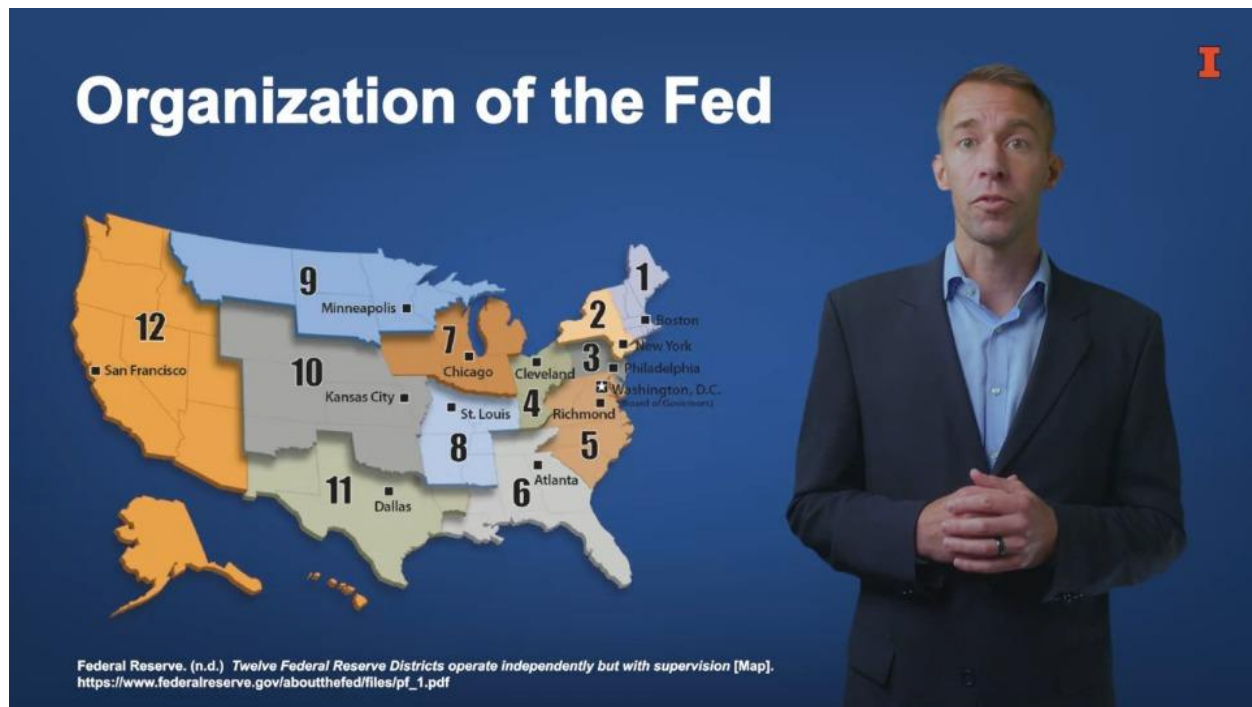
Fed's functions:

1. Control money supply
2. Lender of last resort

What did the Act allow the Federal Reserve System to do? First, it gave the Fed control over the money supply. Second, the Fed could act as a lender of last resort to banks. The lender of last resort function has been used for interventions during times of market-wide stress. This includes the financial crisis of 2008 and the COVID-19 pandemic.



The Federal Reserve Act established three parts of the new central bank, one independent federal agency, the Board of Governors in Washington DC, 12 regional Federal reserve banks, and the Federal Open Market Committee, or FOMC. The board of Governors consists of seven members. The board of governors guides the policies of the entire Federal Reserve System.



The 12 Regional Federal Reserve banks are Boston, New York, Philadelphia, Cleveland, Richmond, Atlanta, Chicago, St. Louis, Minneapolis, Kansas City, Dallas, and San Francisco. The regional Federal Reserve banks were placed across important financial centers. This distributes power of the Fed across the United States.



The reserve banks implement the Federal Reserve policies and function. For instance, they examine and supervise financial institutions, act as lenders of last resort to banks

in the district. They also provide US payment system services for banks in the district. There are 12 members of the Federal Open Market Committee, or FOMC. This always includes the seven members of the Board of governors and the president of the Federal Reserve Bank of New York. The remaining four members come from a rotating group of the remaining Federal Reserve bank presidents. The main task of the FOMC is to set interest rate targets. This is commonly referred to monetary policy.

Summary

Central banks control money supply

Central banks act as lender of last resort

Federal Reserve System
of the United States:

1. Board of Governors
2. 12 Reserve Banks
3. FOMC



What have we learned in this lesson? Before central banks existed, paper money was often issued privately. In this system, the money supply was unpredictable and bank runs occurred frequently. To control the money supply and to ensure the stability of the banking system, central banks were founded. Central banks are also the lender of last resort that will lend cash to banks in times of crisis. In the United States the Federal Reserve System fulfills these functions through three entities. The Board of governors, 12 Federal Reserve banks, and the Federal Open Market Committee.

Lesson 2-1.2. Purpose and Functions of the Federal Reserve

Learning Objectives

Mandate of Federal Reserve System

Five key functions:

1. Monetary policy
2. Financial stability
3. Regulation and supervision
4. Payment system
5. Consumer protection and community development



Hello, and welcome to this lecture on the five key functions of the Federal Reserve System. In this class, we will discuss the mandate of the Federal Reserve System. By law, the Fed exists to promote the effective operation of the US economy. We will see what this means in practice, we will see the five key functions of the Fed. These five key functions are, to conduct monetary policy, promote financial system stability, to regulate and supervise financial institutions, to provide an efficient payment system, and finally, to promote consumer protection and community development. We will discuss these functions one by one and see how they connect with the goals of the Fed.

Fed Function #1: Monetary Policy

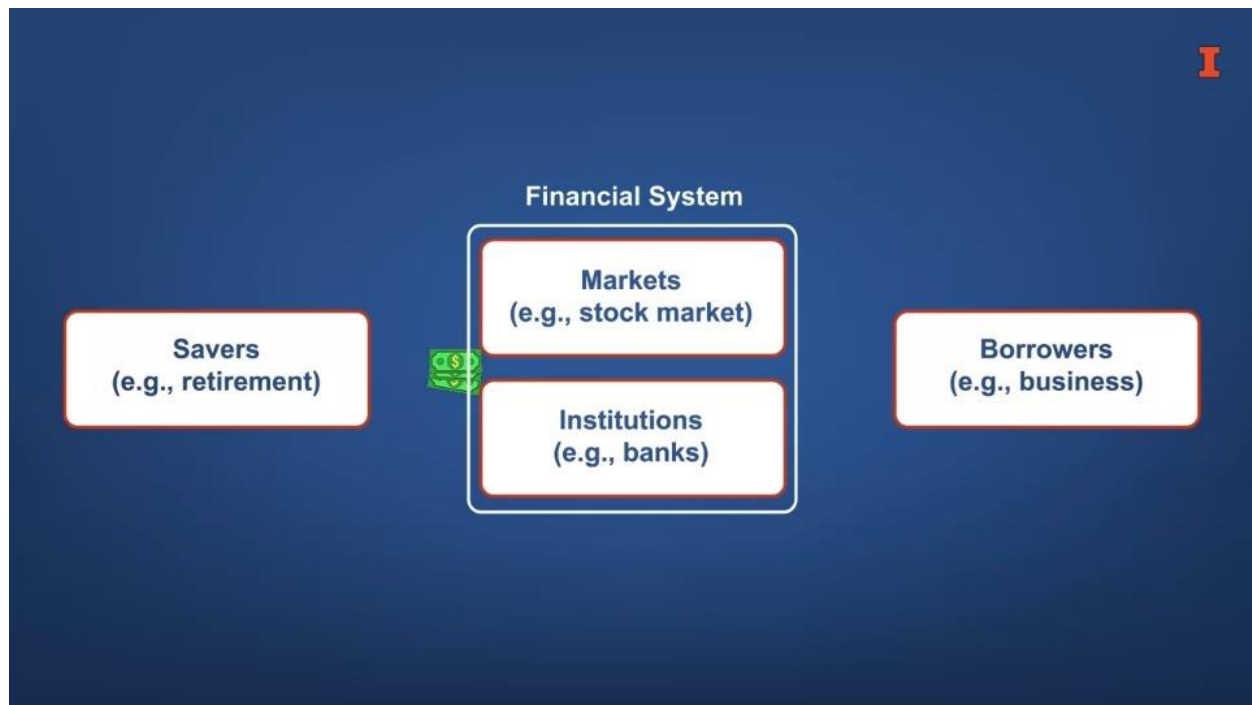
Fed conducts monetary policy

Monetary policy goals:

1. Maximum employment
2. Stable prices
3. Moderate long-term interest rates



Let's get started, the Federal Reserve Act of 1913 created the Federal Reserve System. Under the Act, the Fed must conduct monetary policy to foster a productive and stable economy. Monetary policy has evolved over time, nowadays it involves controlling short term interest rates. By law, the Fed must use monetary policy to effectively promote the goals of maximum employment, stable prices, and moderate long-term interest rates. These are the goals, but what do they mean? Maximum employment usually refers to a situation when most people who are looking for work are gainfully employed. Stable prices encourage savings and investment. By contrast, when inflation is high, asset returns can be lower, this reduces households' incentives to save and businesses incentives to invest. Similarly, moderate long term-interest rates increase business incentives to invest. Under maximum employment, stable prices and reasonable long-term interest rates, the belief is that there will be stable growth and a thriving economy. This is what the Fed hopes to achieve.



The second key function is financial stability, what is financial stability? Generally, we consider a financial system to be stable when it provides services to households and businesses at reasonable cost and without interruption. The financial system consists of financial markets and financial institutions such as banks. A stable system channels funds from savers to borrowers, this allows for an efficient allocation of capital across the economy.

Fed Function #2: Financial Stability

Fed monitors financial system

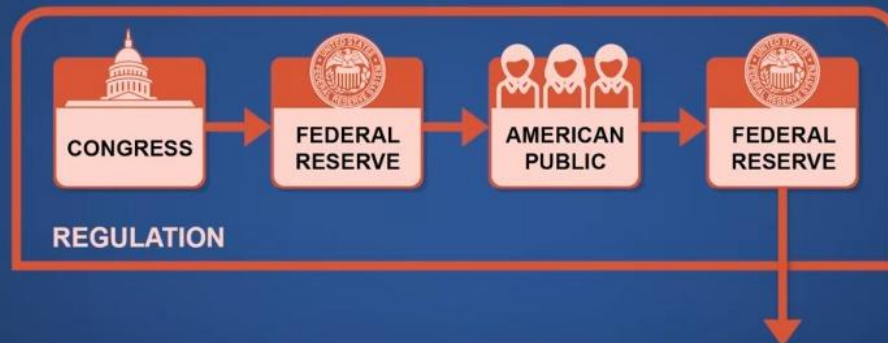
Micro-prudential ⇒ individual firms

Macro-prudential ⇒ system-wide risks



To fulfill this responsibility, the Federal Reserve monitors the financial system. In fact, it monitors using two approaches, the first is called the micro-prudential approach. This focuses on risks at specific institutions, this could be a bank or an insurance company. The micro-prudential approach is related to bank supervision, we will discuss this shortly. The second approach is macro-prudential, this instead looks out for risks that could impact the entire financial system. The Federal Reserve monitors several measures of financial market stress. An example is the amount of borrowing or leverage in the financial system, leverage can amplify losses and cause defaults. Another example is the amount of short-term borrowing by financial firms. Short-term borrowing can cause runs that disrupt the flow of credit to the economy.

Fed Function #3: Regulation & Supervision ^I



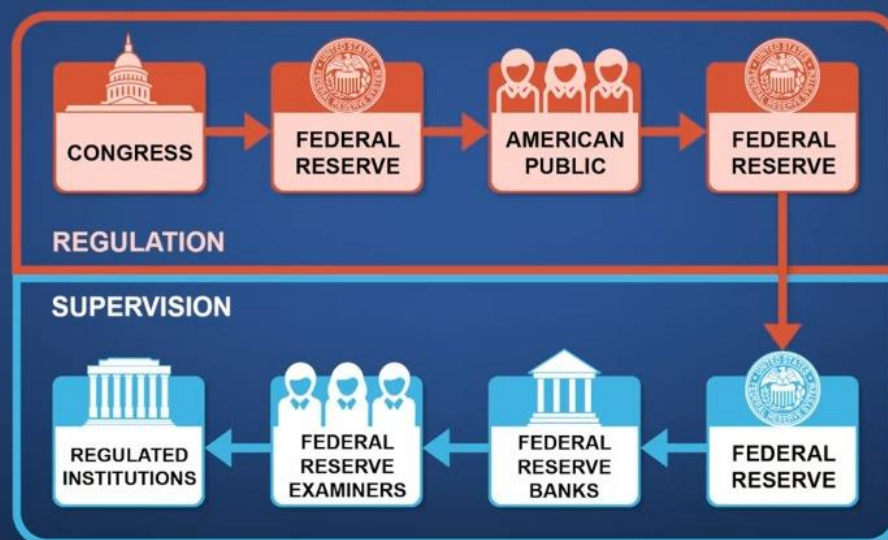
The third key function of the Federal Reserve is to regulate and supervise financial institutions. Regulation and supervision are distinct but complementary activities. Regulation relates to the writing of rules, these rules often place restrictions on financial institutions. This process starts with legislation in Congress, this gives the Federal Reserve the authority to draft rules. With input from the public, the rules are finalized and published, financial institutions must then comply with the rules.

Fed Function #3: Regulation & Supervision



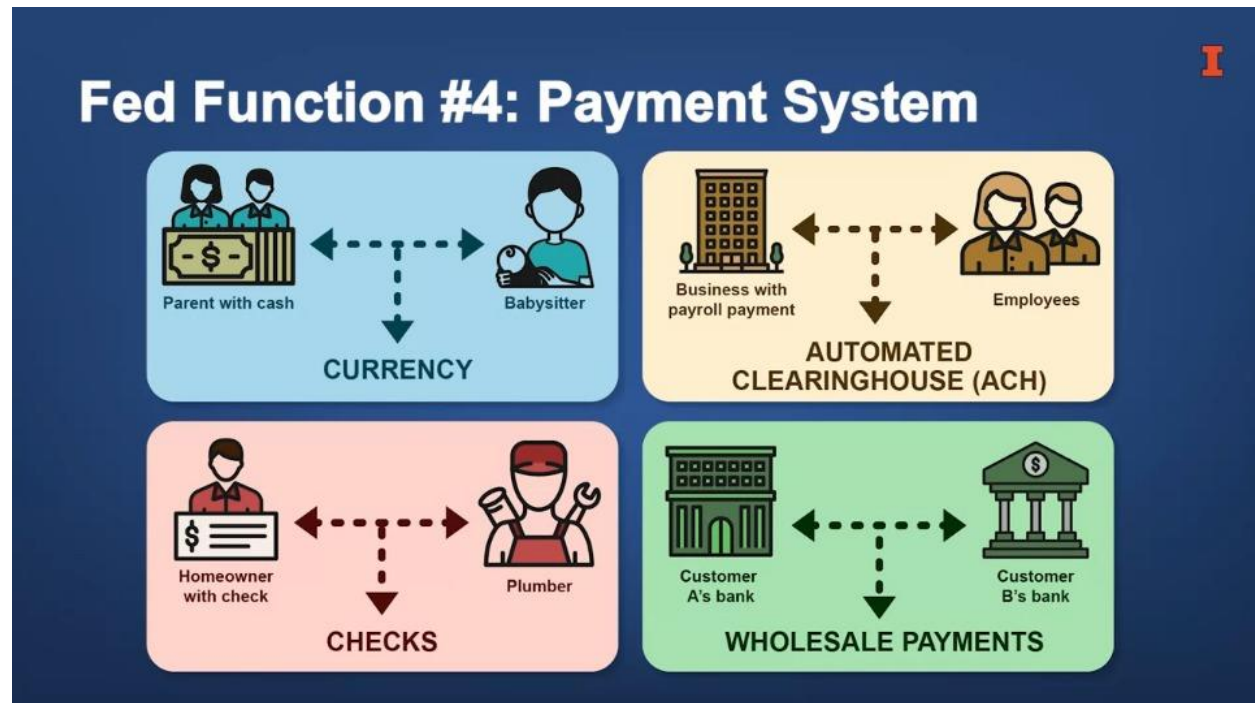
The truth in lending act is a good example, under this rule, the Federal Reserve requires lenders to clearly disclose lending terms and costs to borrowers. This includes the interest rates on a credit card.

Fed Function #3: Regulation & Supervision



Supervision, on the other hand, relates to the compliance with the regulation. For this purpose, the Federal Reserve employs, and trains bank examiners. Examiners conduct visits to regulated financial institutions to ensure compliance with the rules, this involves

in-depth inspections of the bank's operations. If rules are violated, supervisors can impose fines and other penalties such as restrictions on dividend payments. Supervision relates to financial stability; it is a micro-prudential approach. Supervision ensures the safety and soundness of individual financial institutions.



The fourth key function of the Federal Reserve is to provide an efficient payment system. Currently, the Federal Reserve supports a variety of payments, this can be currency. The Federal Reserve issues paper money or cash that traditionally was used for many household transactions. The Federal Reserve also facilitates check clearing. Automated Clearinghouse, or ACH, settles small value bank transfers like wage payments to employees or the payment of your utility bill. And also large value transactions or wholesale payments among businesses and banks.

Fed Function #5: Consumer Protection

Improving fairness and access

Truth in Lending Act

Community Reinvestment Act



The fifth key function of the Federal Reserve is promoting consumer protection and community development. Improving access to affordable financial products provides financial stability for families. To this end, the Federal Reserve implements various consumer protection, fair lending, fair housing, and community reinvestment laws. We mentioned the Truth in Lending Act earlier, this is about transparency and fairness. The Community Reinvestment Act is another important example. Under this rule, the Federal Reserve collects data on how much credit flows to communities and encourages financial institutions to provide access to credit for the entire communities that they serve. This includes lending to low- and moderate-income neighborhoods.

Summary

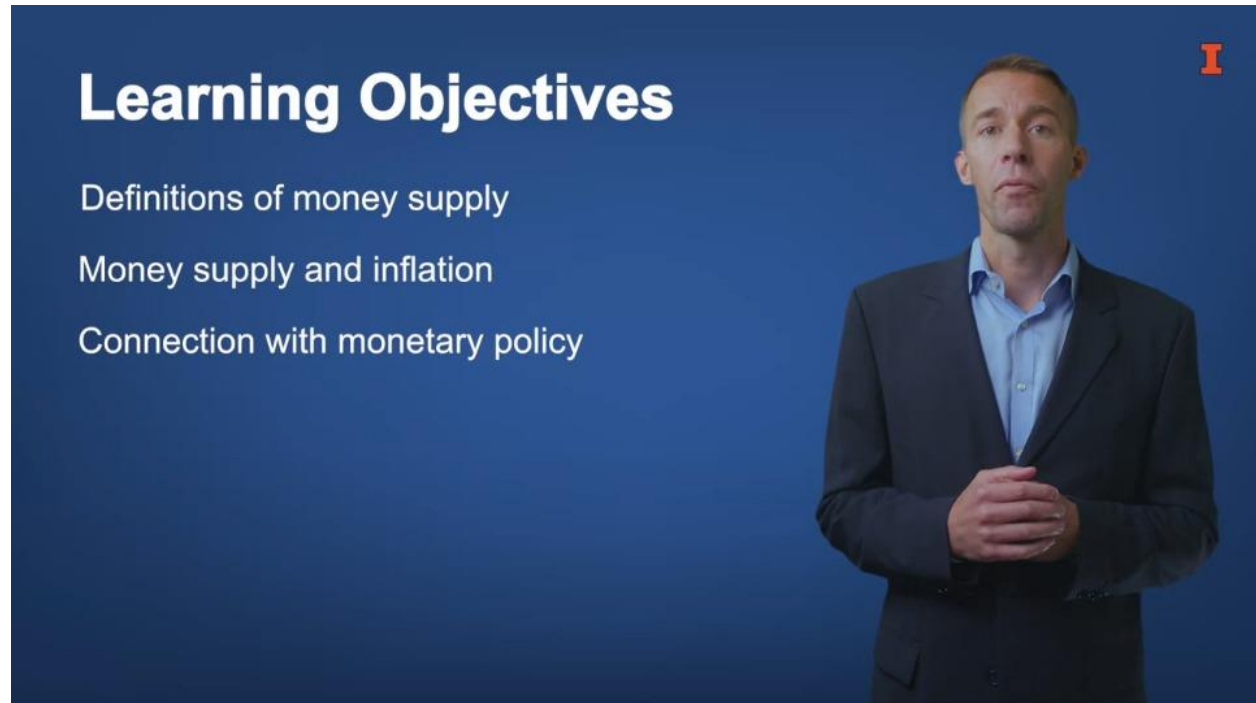
1. Conduct monetary policy
2. Promote financial stability
3. Regulation and supervision
4. Payment system
5. Protect customers and communities



In this session, we have provided a big picture review of the purposes and functions of the Federal Reserve System. We discussed the five key functions of the Fed, conducting monetary policy, financial system stability, regulating and supervising financial institutions, providing and efficient payment systems, and finally, promoting consumer protection and community development. Taken together, these functions support the US economy and more generally, serve the public interests.

Lesson 2-2: Traditional Monetary Policy

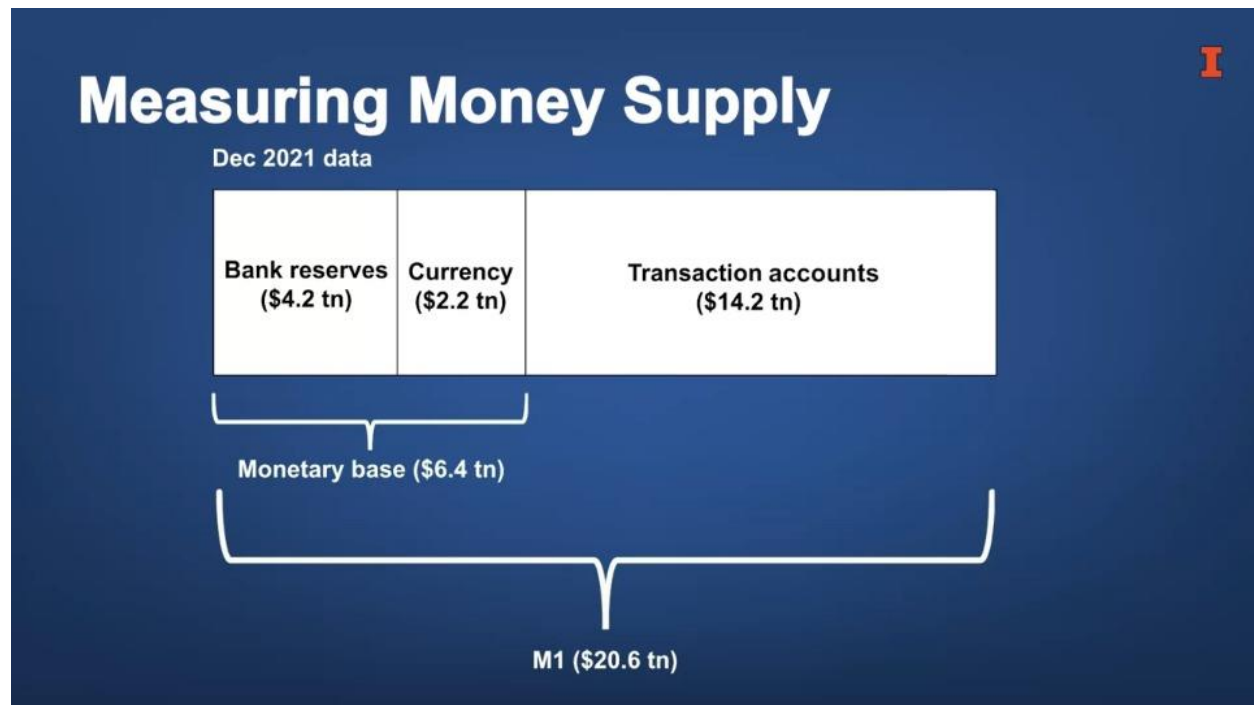
[Lesson 2-2.1. Money Supply and Monetary Policy](#)



Learning Objectives

- Definitions of money supply
- Money supply and inflation
- Connection with monetary policy

Hello, and welcome to this lecture on the money supply. In this class, we will discuss what the money supply is and how it is measured. We will introduce two ideas about the money supply. First, the money multiplier, second, the quantity theory of money. These ideas connect the money supply with important economic outcomes including inflation. If central banks aim to target inflation, then targeting the money supply can make sense. We will discuss how central banks control the money supply. Let's get started. Money is what we use when we buy and sell things. Broadly speaking, it is a reliable means of payment between a buyer and a seller. The money supply is the amount of money in circulation. The money supply includes dollar notes and coins, but it can also include other items as well.



Let's be more precise about how we measure the money supply. The narrowest definition is the monetary base. The monetary base includes the amount of currency in circulation. It also includes bank reserve balances. These are deposits held by banks in their accounts at the Federal Reserves. A somewhat wider definition of money supply is M1. M1 starts with the monetary base, we then add on transaction deposits. This includes our checking account balances at banks. The idea of M1 is to capture cash, and how many loanable funds bank have. These are funds that banks have available to lend out.



But why are loanable funds so important, and how do they relate to the money supply? Consider the following example. Let's say you deposit \$100 at your bank; the bank keeps 10% or \$10 at the Federal Reserve as a reserve. This could be required or simply prudence on the part of the bank. It lends out the remaining \$90. The \$90 will appear as a new deposit in the borrower's checking account. Again, the bank keeps 10% or \$9 at the Federal Reserve as a reserve and lends out 81. Now, the 81 will appear as a new deposit. This process could continue and on, and it is often called the money multiplier. Each time funds are loaned out, bank deposits, and therefore, the money supply will increase. In fact, the 10% reserve is what limits the money supply.

Measuring money supply

Monetary base = reserves + currency

M1 = Base + transaction accounts

M2 = M1 + savings deposits +
small time deposits + retail MMMF shares

M3 = M2 + large time deposits +
institutional MMMF shares + repos +
larger liquid assets



Let's look at two more measures of money supply. The most common definition is M2. This is M1 plus other items that can be converted into cash at a relatively short notice. This includes savings deposits, small time deposits under \$100,000, and retail money market mutual fund shares. The widest definition is M3. This takes M2, and adds large time deposits, institutional money market funds, short-term repurchase agreements, and larger liquid assets.

Money supply and inflation

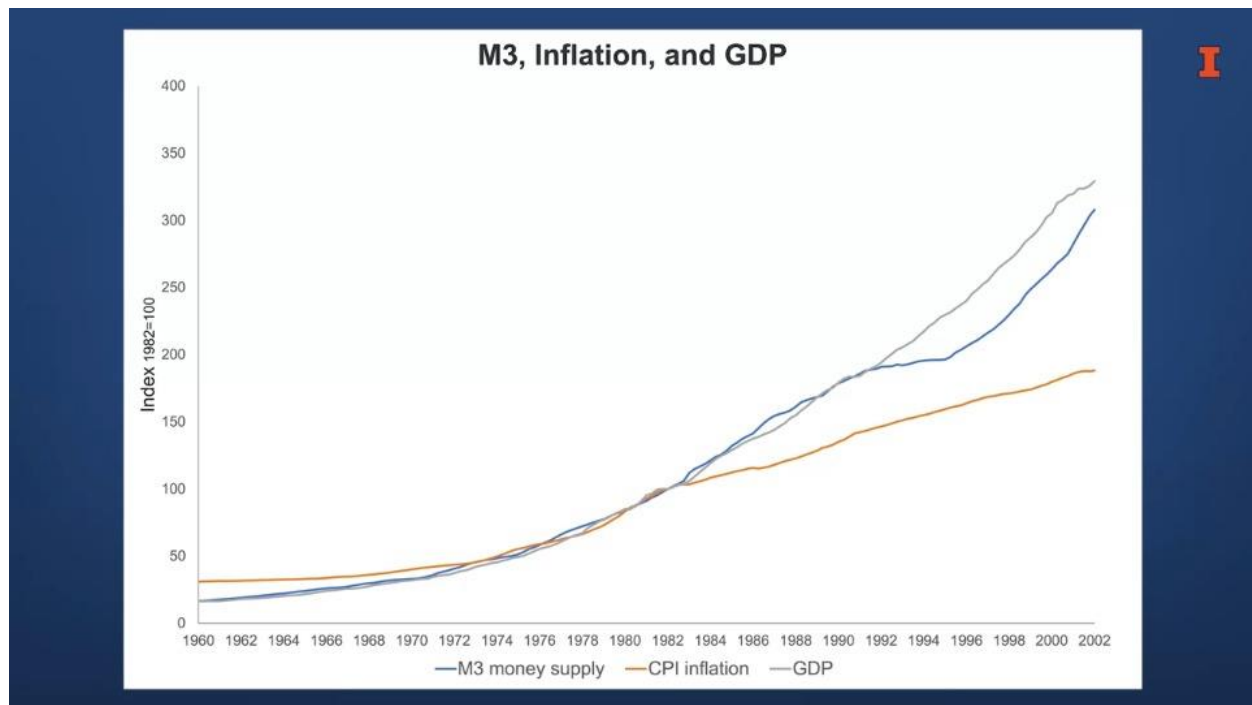
Historically, money supply growth $\Rightarrow$ inflation

“Quantity Theory of Money”

Idea: control money supply $\Rightarrow$ control inflation



Historically, measures of the money supply were closely related to key macroeconomic outcomes. Past episodes of high inflation coincide with periods when the money supply has grown fast. Why? Well, like any other commodity, the value of money declines if there is excess money in circulation. The relationship between money supply, inflation, and economic growth is called the quantity theory of money. It is one of the oldest ideas in economics and was popularized by Milton Friedman of the University of Chicago. Why does this relationship matter? Well, central bank's target stable prices and maximum employment. This is their mandate. Historically, measures of the money supply were closely related to inflation and economic growth. For this reason, central banks have used measures of the money supply when deciding on monetary policy.



Let's look at this in the data. This is the US context going back to the 1960s. Here we have the M3 money supply. Now we add the consumer price index to measure inflation. You can see that M3 money supply tracks inflation reasonably well through some divergence after the recession in the 1980s. Similarly, M3 tracks nominal gross domestic product well through until 2000.

Money supply and monetary policy

Idea: control money supply $\Rightarrow$ control inflation

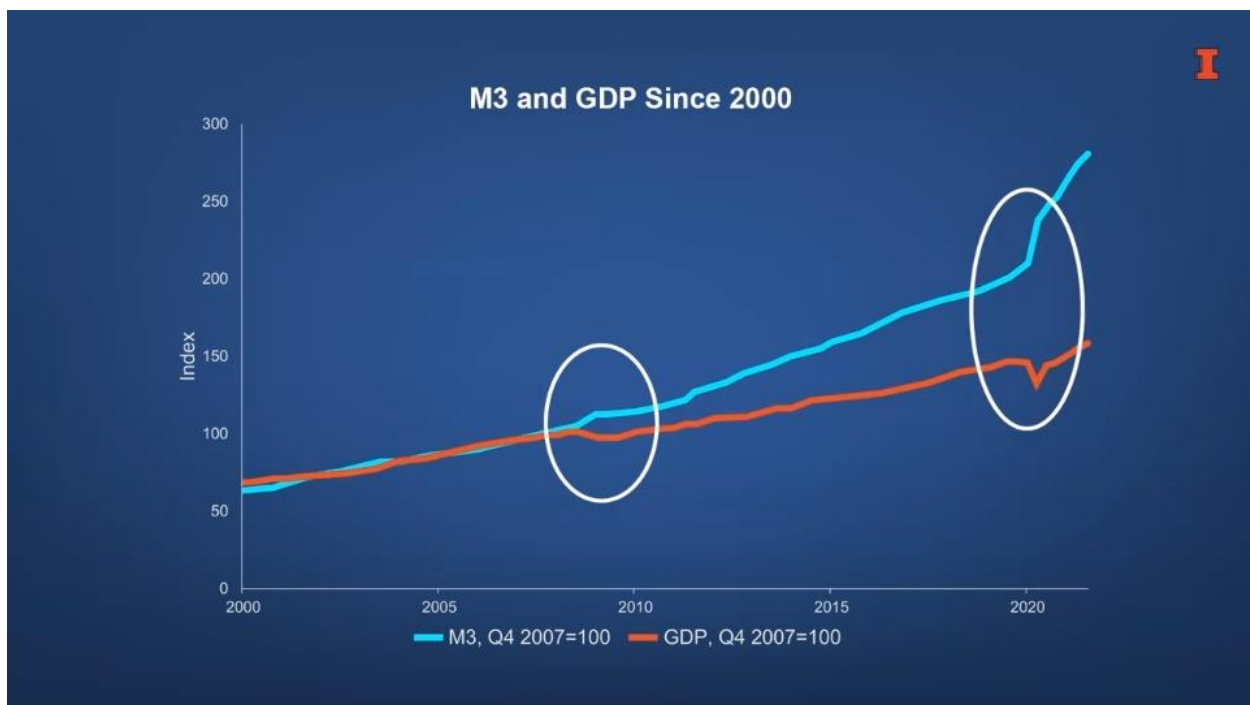
How? Target banks with:

1. Reserve requirements
2. Open market operations



In principle, the Federal Reserve could control the money supply to achieve its

mandate. In practice, they could also do this by targeting banks in a couple of ways. First, the federal reserve can change the reserve requirement for banks. Recall the money multiplier, the higher the reserve requirement is, the more reserves banks must hold, and the less loans banks can make. In other words, the Federal Reserve can control the money multiplier. Second, the Federal Reserve can conduct open market operations. In open market operations, the Fed trades securities with banks, treasuries, for example. If the Federal Reserve sells securities to a bank, the bank must pay the Fed using loanable funds. As a result, the bank will have fewer funds to make loans, which will decrease the money multiplier and the money supplier. Are we done? Not quite. Unfortunately, as time went on, the relationship between money supply, inflation, and GDP became less reliable in the data. First, there are now new payment technologies and ways for households to store cash. This made it harder to measure money supply. The federal reserve has stopped circulating M3.



Second, the relationship between M3 and inflation or GDP have not been stable since the year 2000. Since the financial crisis of 2008, the relationship between M3 and GDP has broken down. You can see the clear divergence here. It happened again during the COVID-19 pandemic. Most central banks now regard money supply measures as less important when making monetary policy decisions.

Summary

Measurement of money supply:
monetary base, M1, M2, M3

“Money multiplier”

Relationship between money,
inflation, and GDP

Monetary policy can affect money supply



In the session, we have discussed the concept of the money supply and its connection with monetary policies. The key takeaways are, first, we can measure the money supply several ways. We discussed the monetary base M1, M2 and M3. Second, the money multiplier connects bank reserves the money supply. Third, we examined the relationship between money supply, inflation, and GDP. Fourth, we saw that monetary policy can increase or decrease the money supply.

Lesson 2-2.2. The Federal Reserve's Balance Sheet

Learning Objectives

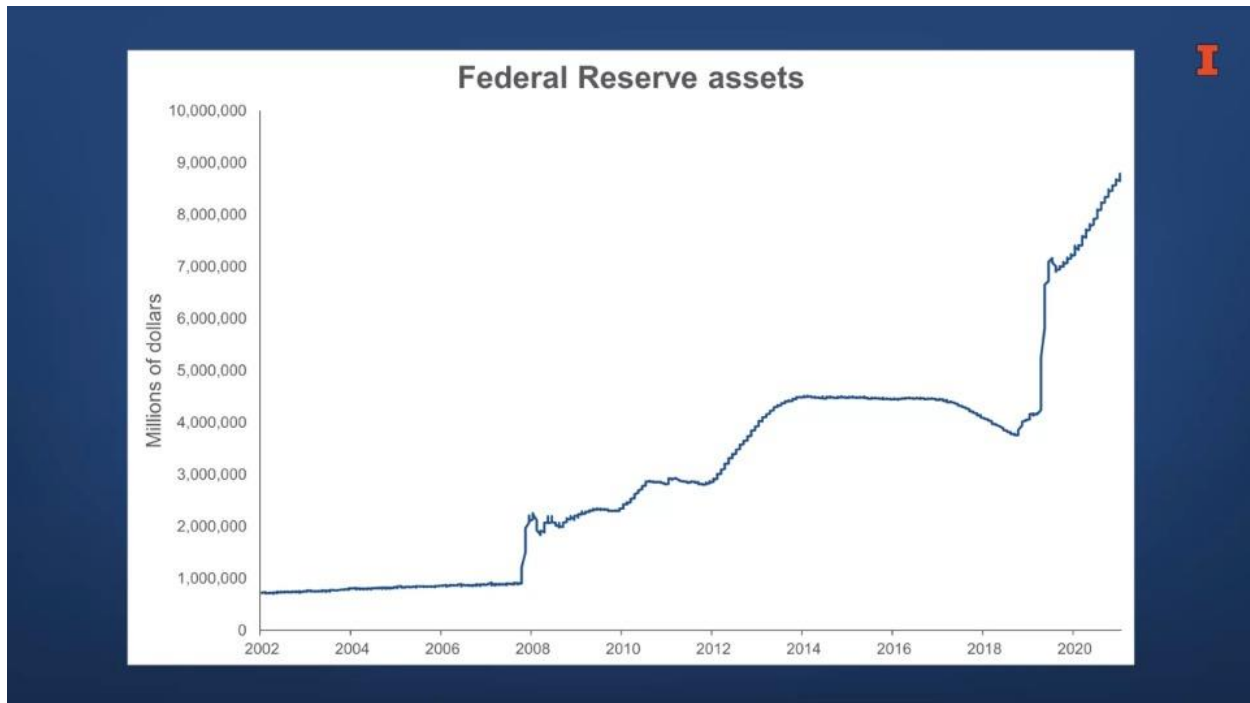
Fed's assets and liabilities

Monetary policy actions ⇒ balance sheet

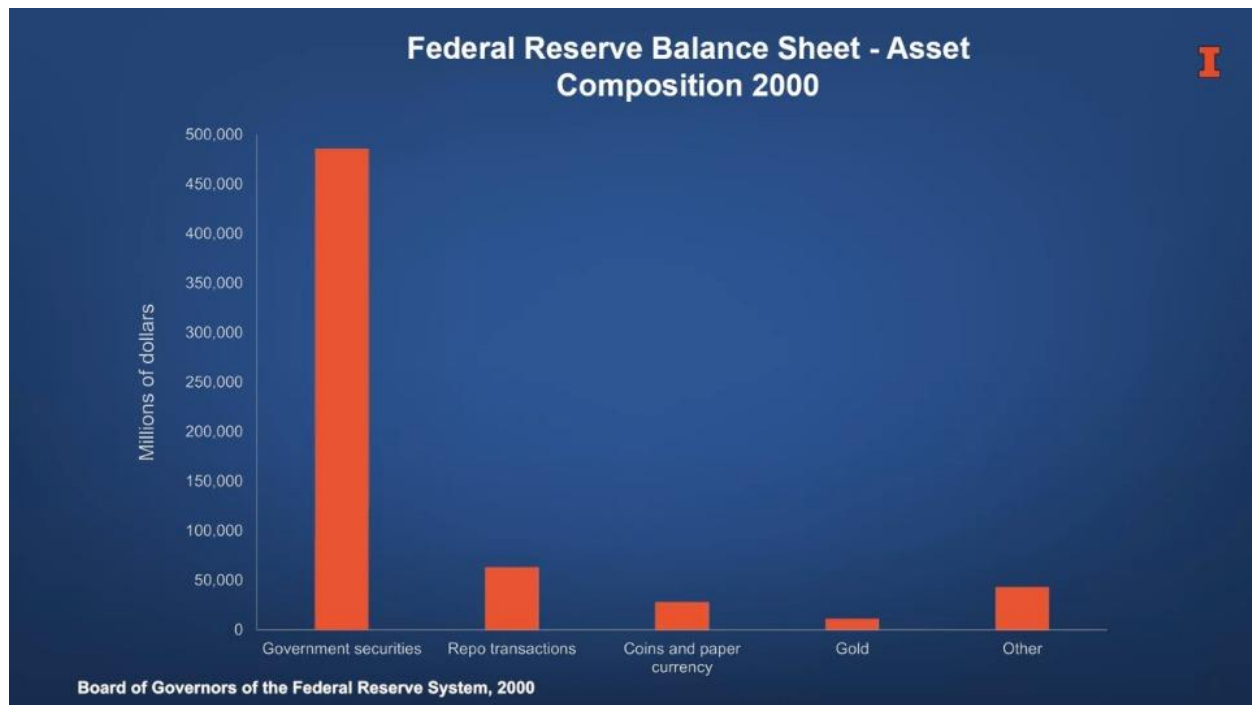
Dramatic changes in recent years



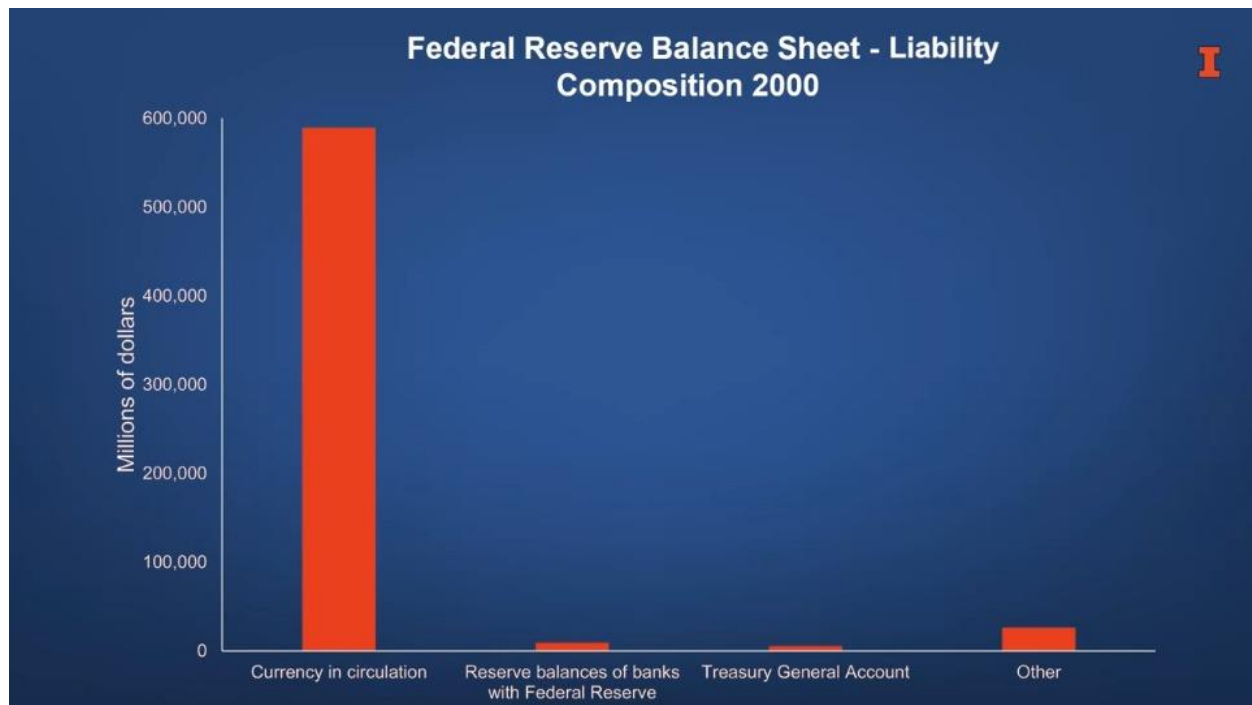
Hello and welcome to this class on the Federal Reserve balance sheet. The Federal Reserve system is the Central Bank of the United States. Like any financial institution, the Fed has a balance sheet. Like any balance sheet, there will be assets and there will be liabilities. What's different is that the Fed uses its balance sheet to achieve its monetary policy goals. For example, when the Fed decides to buy assets and expand its balance sheets, it is a monetary policy action. We will take a close look at the Federal Reserve balance sheet. We will look at its size. We will examine its components. What are the main assets? How are these assets funded? Then we will see how the composition of the Fed's balance sheet has changed in the recent years. We will see how dramatic monetary policy actions in response to crisis have led to these dramatic changes. This will be an important theme of this class. Let's get started. Every week, the Federal Reserve publishes a detailed balance sheet. This information is made available online and market participants pay close attention.



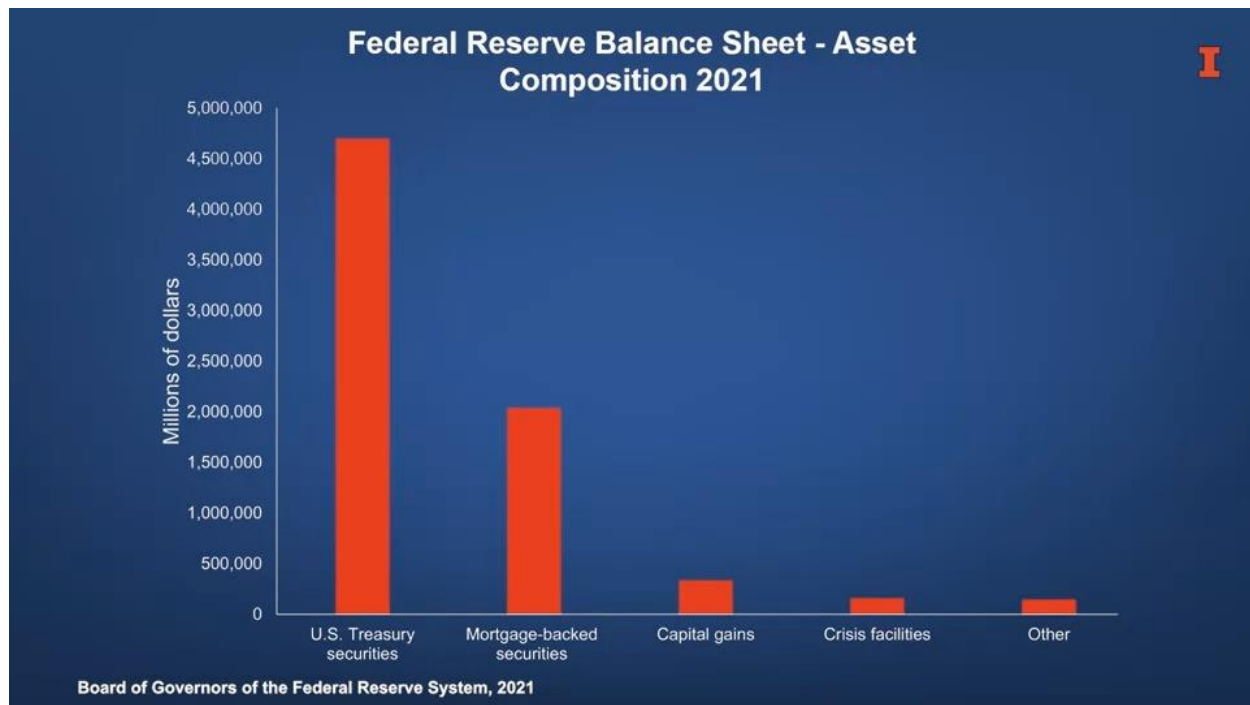
Let's start by looking at the size of the Federal Reserve balance sheet over time. By size, I mean the total assets of the Federal Reserve. As you can see, in the early 2000, the Federal Reserve held assets worth about \$800 billion dollars. With the 2008 financial crisis, the balance sheet more than doubled. At this time, the Federal Reserve conducted some major monetary policy interventions. We will discuss these interventions in another class, but you can see right away that they had a traumatic effect. By 2009, the Federal Reserve held assets worth a little over two trillion. From 2009-2014 the balance sheet more than doubled again to over four trillion dollars. Finally, during the COVID-19 pandemic, the balance sheet size doubled yet again and reached eight trillion by the summer of 2021. What is the key takeaway here? There has been an explosive growth in the Federal Reserve balance sheets since 2008. This growth reflects monetary policy actions.



Now let's look at a breakdown of the most important assets. We will start with the pre-2008 balance sheet. This will tell us something about the Fed's operation in normal times. On Wednesday, January 12, 2000, the Federal Reserve held assets worth about 630 billion. By far, the largest asset position, accounting for three quarters of all assets, is Government securities worth 485 billion. The second largest positions are report transactions. This is when the Federal Reserve make cash loans on a short-term basis. This is usually part of an open market operation which we will discuss in another class. The Federal Reserve also holds currency which includes coin and paper currency and gold.



Now let's look at the liability side of the balance sheet. Here the largest position is currency in circulation. That is, the dollar notes and coins in your pocket worth 590 billion. Then there are reserve balances of banks with the Federal Reserve. These include both required and access reserves. Remember these first two liabilities, currency plus reserves are together called the monetary base. The monetary base is a liability of the Fed. It is issued by the Fed and backed by assets of the Federal Reserve system. The Treasury General Account is the checking account of the US government. All US government payments are made from this account. Tax receipts are also deposited in this account. The balance sheet on January 12th, 2000, was small compared to 20 years later. Indeed, in the early 2000, it was not even a 10th of the size of the balance sheet during the COVID-19 pandemic. How did this expansion of the balance sheet happen?



Let's examine the asset side first. This is the data for January 7th, 2021. There are a couple of things to note for the 2021 data. First, the amount of assets held by the Federal Reserve was much bigger. The total across all assets is 7.3 trillion. Remember, it was only 630 billion before. Second, the list of items has changed, too. Now we see mortgage-backed securities and crisis facilities showing up. The largest item is still US treasury securities worth over 4.6 trillion. In addition, the Federal Reserve held mortgage-backed securities. That is bonds backed by mortgages from government sponsored enterprises worth over two trillion. These bonds are guaranteed by the US government. Government securities accounted for over 90% of the balance sheet. Capital gains are next. These are profits on securities that have not been sold yet. These are also called unrealized capital gains. Then we see crisis facilities. These are various loan programs. They are mostly to banks and funds were provided during times of market wide stress. Banks will have to repay these loans. What is the main driver for asset growth? Clearly, it is the government and government-backed securities held by the Federal Reserve.

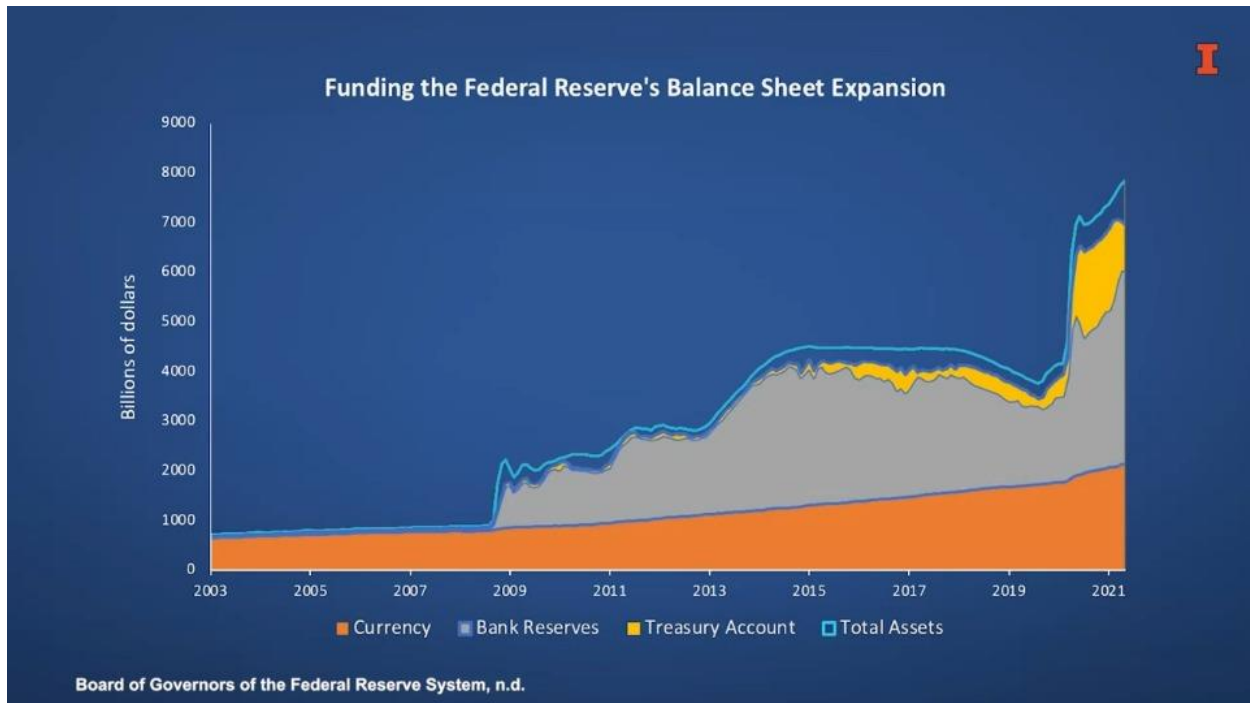


How did the Federal Reserve finance this massive balance sheet expansion? A regular business or household might have to borrow to expand. However, the FED controls the money supply, so it could do this instead.



Did the Federal Reserve print more paper money? Let's look at total currency in circulation over the same period. As you can see, currency in circulation has increased a lot. Over the span of 20 years, it almost quadrupled. However, this increase isn't the

whole story. If the Federal Reserve did not print paper currency, what did they do instead? The Federal Reserve does not buy government securities from the US Treasury directly but purchases them from banks. How did the FED pay these banks? Not with paper currency. Instead, they are paid with digital money. This takes the form of a credit, to the banks' reserve accounts at the FED.



The FED increased the amount of banks reserves in the system. As you can see, bank reserves at the Federal Reserve expanded to about 40% of liabilities by 2021. What's missing here? Most of the remaining liabilities belong to the US government general account about 20% at its peak. Why? Well, the US government had a bigger footprint after 2008 and especially in 2020. This resulted in more money being deposited at the Federal Reserve. The government used these deposits to pay for various fiscal programs.

Summary

Balance sheet of the Federal Reserve

1. Assets = government securities
2. Liabilities = currency + bank reserves

Dramatic expansion due to crisis interventions



Let's wrap up. In this lecture, we have discussed the Federal Reserve balance sheets and its components. The key takeaways are first, the main asset held by the Federal Reserve is government securities. Second, the main liability is the monetary base. This consists of currency in circulation and bank reserves. Third, the size of the balance sheet went up by a factor of 10 between 2008 and 2020. At this time, the FED used its monetary policy tools to support the economy. It paid for this support by increasing the monetary base, particularly the amount of bank reserves in the financial system.

Lesson 2-2.3. Open Market Operations

Learning Objectives

Define open market operation (OMO)

Examples of OMOs

OMO $\Rightarrow$ bank reserves $\Rightarrow$
federal funds rate (monetary policy)



Hello, and welcome to this class on open market operations. Open market operations are central to the monetary policy toolkit. If the Federal Reserve decides to increase interest rates, then this could be achieved using an open market operation. What is an open market operation? We will see that an open market operation involves the sale or purchase of securities by a central bank. We will also see how this works in practice. We will discuss how open market operations directly affect the quantity of reserves in the federal funds market. We will see how these open market operations determine the equilibrium interest rate in this market. The federal funds rate, remember, the federal funds rate is the main target interest rate of monetary policy in the US.

Monetary Policy

Federal Reserve $\Rightarrow$ target interest rate

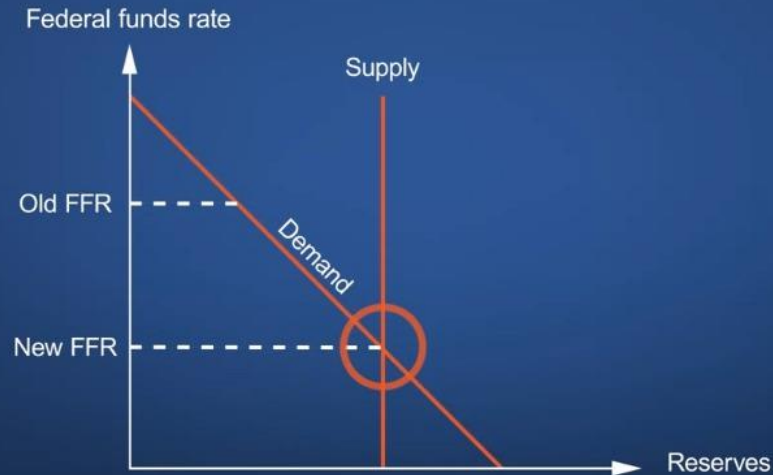
Target rate = federal funds rate (FFR)

Open market operation (OMO) $\Rightarrow$ FFR



Let's get started, monetary policy involves choosing a target interest rate. This target interest rate will hopefully have an impact on the economy. This will allow the Federal Reserve to achieve its goals. The target interest rate of the Federal Reserve is called the federal funds rate, this is a short-term interest rate. In fact, this is the interest rate in the interbank market. Remember, the US interbank market is called the federal funds market. All banks have reserve accounts at the central bank, these reserves are called federal funds. An open market operation is designed to influence the federal funds rate.

Interbank Market



Roughly speaking, how does this work? Well, the Federal Reserve does not change the interest rate directly. Instead, by conducting an open market operation, the fed changes the supply of reserves. The intersection of supply and demand determines the price of these reserves. That price is the same as the interest rate in this market, it is called the federal funds rate. Let's assume the Federal Reserve wants to decrease the interest rate. To achieve this, the Federal Reserve could shift the supply of reserves to the right. The new equilibrium features more federal funds, which will be traded at a lower interest rate.

Open Market Operation

OMO to increase bank reserves

1. Buy securities from bank
2. Fed pays with reserves

Higher reserves $\Rightarrow$ lower FFR



Open market operations allow the Federal Reserve to manipulate the quantity of reserves in the system. How does the Fed do this? Let's continue our example, say the Fed wants to increase bank reserves. They could do this by giving banks more reserves in exchange for some securities. The bank would already own these securities, and the Fed would have to pay a fair price. One more time, if the Federal Reserve wants to lower the federal funds rate, it can buy securities from banks. These purchases are paid for with reserves. Hence, the overall available funds in the federal funds market will increase. This shifts the supply curve to the right. The new equilibrium has more federal funds and a lower interest rate. Let's have a look how this works in practice, there are several steps.



Step 1, the Federal Reserve's Open Market Committee, or FOMC, decides on the target interest rate. Step 2, by buying and selling itself, is relegated to the Federal Reserve Bank of New York. They manage the Federal Reserve system open market account, or SOMA. Step 3 the Federal Reserve Bank of New York's trading desk then purchases securities from the so-called primary dealers. These dealers have reserve accounts at the Federal Reserves. These dealers include the largest banks. What types of securities are usually purchased? Mostly US treasuries. Step 4, when the trading desk completes a purchase, the Federal Reserve's balance sheet expands. On the asset side, we add the purchased securities, and on the liability side, reserves increase. These are new reserves that the Federal Reserve creates. Remember, the Fed controls the monetary base, so it is free to create these reserves. Here's an example of an open market purchase made by the Federal Reserve.

System Open Market Account (SOMA)

Outright Treasury Securities Operations			7/19/2021
RESULTS		AMOUNT (\$Millions)	
Operation Type		Par Submitted	Par Accepted
Outright Coupon Purchase		3,099	1,401
Operation Date:	Monday, July 19, 2021		
Release Time:	10:10 AM		
Close Time:	10:30 AM		
Settlement Date:	July 20, 2021		
Maturity/Call Date Range:	February 15, 2036 - November 15, 2043		
Recent Operations Historical Search			

Federal Reserve Bank of New York, n.d.

On July 19, 2021, the Fed bought us treasury securities with a par value of 1.4 billion. The securities had a remaining maturity of between 15 and 22 years.

System Open Market Account (SOMA)

Domestic Holdings: Includes CUSIP-level data and SOMA Historical Data Export Builder. Data are updated Thursdays with previous day's holdings.		7/15/2021
SECURITY TYPE	VOLUME (\$Thousands)	
US Treasury Bills (T-Bills)	326,044,000.0	
US Treasury Notes and Bonds	4,460,904,892.7	
US Treasury Floating Rate Notes (FRN)	23,846,691.1	
US Treasury Inflation-Protected Securities (TIPS)*	354,886,562.0	
Federal Agency Debt Securities**	2,347,000.0	
Agency Mortgage-Backed Securities***	2,392,591,797.9	
Agency Commercial Mortgage-Backed Securities***	9,769,885.7	
Total SOMA Holdings	7,570,390,829.3	
Change From Prior Week	102,527,270.7	

Federal Reserve Bank of New York, n.d.

Here's a snapshot of the system's open market account portfolio from July 19th, 2021. These were the domestic security holdings of the Federal Reserve system. At this point in time, all the securities are US government debt or debt guaranteed by the US government. The portfolio increased by about 100 billion relatives to the prior week.

OMOs in Recent Years

During 2008 financial crisis ⇒
interest rates near zero

After 2008 ⇒ supply of bank reserves too high

Traditional OMOs ineffective

New approaches needed



The 2008 financial crisis marks a breakpoint in the way open market operations were conducted. The Fed reduced its target for the federal funds from five and a quarter percent in mid 2007 to a range of zero to a quarter percent in December 2008. With short term interest rates close to zero, simply buying more securities to increase reserves was no longer effective. New approaches were needed to stimulate a struggling economy. After the financial crisis, the aggregate supply of banked reserves was now larger than ever. This was a direct consequence of large-scale interventions by the Federal Reserve. This meant that open market operations would no longer be as effective in controlling the short-term interest rate. The Fed needed new monetary policy tools to deal with these problems. We will discuss these issues in a separate lecture.

Summary

OMOs affect Fed's target interest rate via interbank market

Before 2008 crisis, OMOs were main monetary policy tool

After 2008, new tools needed



In this session, we have discussed open market operations and monetary policy. The key takeaways are first, open market operations affect the supply of reserves. This, in turn, determines the federal funds rate, the Federal Reserve's target interest rate. Second, before the 2008 financial crisis, open market operations were the main monetary policy tool. Third, after 2008, with the supply of reserve have been elevated, traditional open market operations have been less important.

Lesson 2-2.4. The Discount Window

Learning Objectives

What is the discount window?

Primary, secondary, and seasonal credit

Monetary policy: discount rate
versus federal funds rate



Hello, and welcome to this lecture on the Federal Reserve's Discount window. We will discuss how the federal reserve uses the discount window to provide a variety of short-term cash loans to banks. We will examine primary, secondary, and seasonal credit. We will see what these loans look like and understand why they exist. We will then draw a connection between the discount window lending and monetary policy, we will understand how the discount rate influences the federal reserve target interest rate. Okay, let's get started. Discount lending has long been a tradition in banking. The basic idea is that the lender first calculates the required amount of interest and fees. The lender then discounts this amount from the face value of the loan. The borrower gets the discounted loan today, this loan amount is already reduced upfront to account for interest and fees, but then the borrower must repay the full-face amount at maturity.



Let's go through a quick example. You want to take out a discount loan with a face value of \$100, the maturity of the loan is one year. Let's say the lender calculates a discount rate of 10%, this covers the interest on the loan and fees. The loan amount that you receive today is \$90, this is the face value of \$100 minus the \$10 of interest and fees, but in a year, you will have to repay the full \$100.

Discount Loan

- Cost of loan paid upfront
- Single payment at maturity
- Short-term loan

A man with short brown hair, wearing a dark blue suit jacket over a light blue button-down shirt, stands with his hands clasped in front of him. He is looking directly at the camera. The background is a solid dark blue. In the top right corner of the slide, there is a small orange 'I' logo.

Why is this type of loan useful? Well, the cost of a loan is accounted for upfront, this

means only one payment is required rather than a series of monthly interest payments. For that reason, discount loans often help with short-term funding needs.

Discount Loans by Federal Reserve

Discount window loans since 1913

Why? Temporary cash needs

Nowadays, secured loans rather
than unsecured discount loan



Federal Reserve Bank of St. Louis. (1967). *Credit discount window at the St. Louis Fed* [Photograph].
<https://fraser.stlouisfed.org/archival/5182/item/527398>

Since 1913, the federal reserve has offered discount loans to banks. Why does the Fed do this? Well, the loans provide cash to banks on short notice, this ensures that there is enough cash in the banking system. In the extreme, this can prevent bank runs and banking panics. Discount loans were provided at a special window at each of the regional reserve banks called the discount window, hence the name discount window lending. Nowadays, the Fed provides short-term cash loans secured by eligible collateral, these are secured discount loans. However, the term discount window stuck, it is still used when the federal reserve lends directly to banks.

Primary Credit Discount Loans

Well-capitalized banks only

Max 90 days

Usually, FFR + markup



The screenshot shows the Federal Reserve's Discount Window website. The header includes the Federal Reserve logo and the text "THE FEDERAL RESERVE Discount Window | Payment System Risk". The date "As of Aug 30, 2023" is displayed. Below the header is a search bar. The main content area is titled "Current Interest Rates" and contains a table with four rows: Primary Credit (5.50%), Secondary Credit (6.00%), Seasonal Credit (5.40%), and Fed Funds Target (5.25 - 5.50%). Below this table is a section for the "Bank Term Funding Program" with a rate of 5.50% as of August 30, 2023.

Current Interest Rates	
Primary Credit	5.50%
Secondary Credit	6.00%
Seasonal Credit	5.40%
Fed Funds Target	5.25 - 5.50%

Bank Term Funding Program
5.50%
August 30, 2023

At the federal reserve, banks currently have access to several types of discount loan lending. We will focus on primary credit; this is the most important category. The main purpose of primary credit is to provide cash in case other funding is not available. Primary credit discount loans are available to well-capitalized healthy banks. These are short-term loans with a maximum maturity of 90 days. The interest rate on primary credit loans is set relative to the target federal funds rate. This is usually higher than the interest rate on other forms of short-term credit, this incentivizes banks to borrow in the private cash markets. The Fed should be a last resort.

Primary Credit Discount Loans

Well-capitalized banks only
Max 90 days
Usually, FFR + markup
Secured by eligible collateral

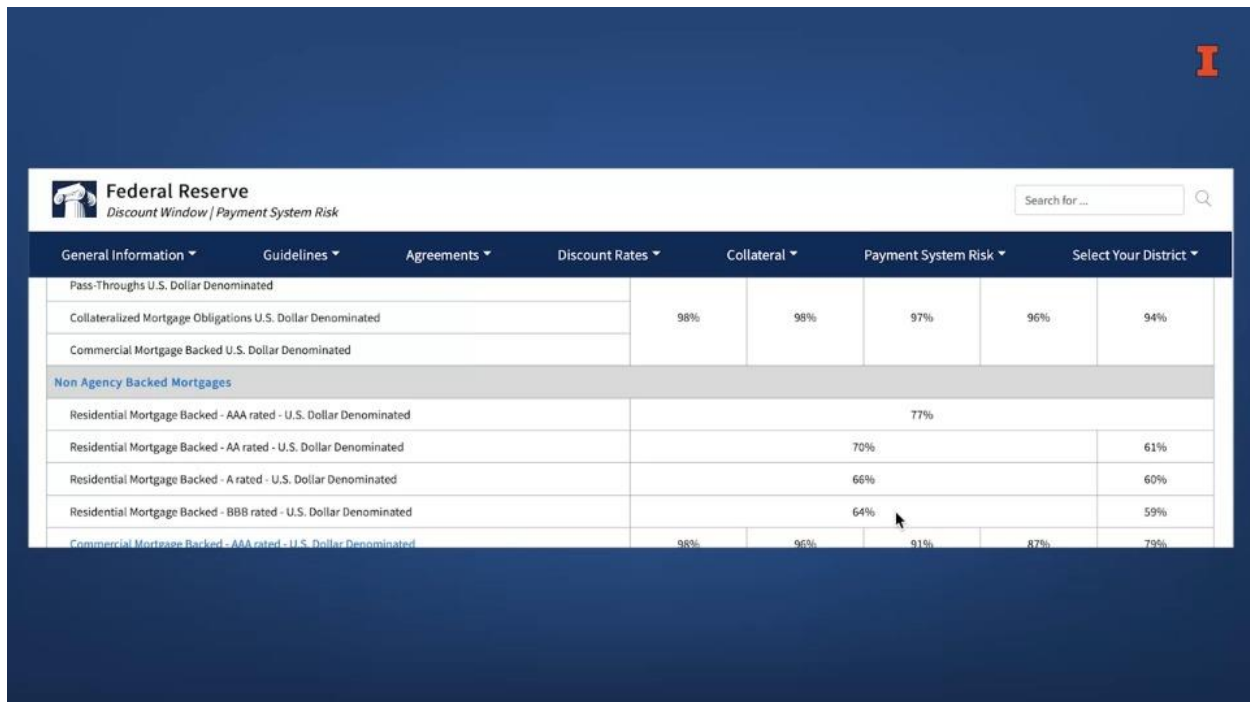


These are secured loans. The federal reserve accepts a broad range of collateral, including commercial and industrial loans, consumer loans, residential and commercial real estate loans and US treasuries. Some of these types of collateral are riskier than others. To account for risk, the Fed applies margins indicating how much a bank can borrow against each type of collateral.

Federal Reserve Discount Window Payment System Risk					
Effective Date: March 14, 2022					
Securities	Margins for Securities ¹ (% of market value)				
	Duration Buckets				
	0-1	>1-3	>3-5	>5-10	>10
U.S. Treasuries & Fully Guaranteed Agencies					
Bills, Notes, Bonds, Floating Rate Notes, and Inflation-Indexed	99%	99%	98%	97%	95%
STRIPS	96%				92%

Here's an example of a margin table, these margins are as a percentage of the market

value. A US treasury with a remaining maturity of one year is very safe. A bank can borrow 99 cents on the dollar. In contrast, a loan for commercial real estate with a remaining maturity of one year is still rather risky.



General Information	Guidelines	Agreements	Discount Rates	Collateral	Payment System Risk	Select Your District
Pass-Throughs U.S. Dollar Denominated						
Collateralized Mortgage Obligations U.S. Dollar Denominated			98%	98%	97%	96%
Commercial Mortgage Backed U.S. Dollar Denominated						94%
Non Agency Backed Mortgages						
Residential Mortgage Backed - AAA rated - U.S. Dollar Denominated					77%	
Residential Mortgage Backed - AA rated - U.S. Dollar Denominated					70%	61%
Residential Mortgage Backed - A rated - U.S. Dollar Denominated					66%	60%
Residential Mortgage Backed - BBB rated - U.S. Dollar Denominated					64%	59%
Commercial Mortgage Backed - AAA rated - U.S. Dollar Denominated			98%	96%	91%	87%

Banks can only borrow between 35% and 95% of the value of the loan.



Discount Window and Monetary Policy

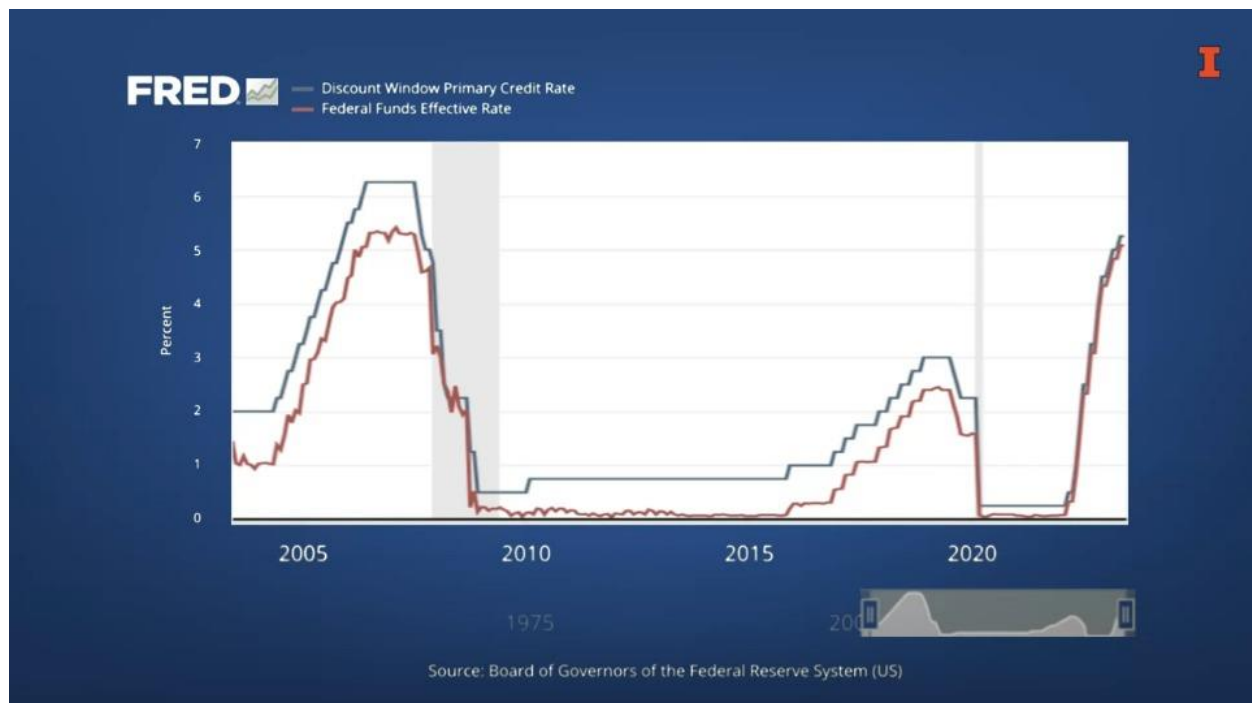
Discount loan substitute for interbank loan

Fed sets: discount rate > target FFR

Result: discount rate > realized FFR

How is the discount window related to monetary policy? The discount window helps the

federal reserve to control short-term interest rates in the economy, how does that work? Well, there are two things to keep in mind. First, a discount loan from the Fed or an interbank loan from another bank are close substitutes, both are short-term loans. Any healthy bank can access either the interbank market or the discount window. Second, recall that the discount rate is set near the upper bound of the federal fund target range. Now, what happens if the realized interest rate on the federal funds exceeds the discount rate? Well, it would be cheaper for a bank to borrow from the discount window. This shift in demand away from the interbank loans will put downward pressure on the federal funds rate. Hence, the discount rate enforces the upper bound of the federal funds target set by the central bank. It ensures the realized federal funds rate does not get too high.



We can see this in the data. From 2003 onwards, the primary credit rate always sits above the federal funds rate.

Summary

Discount window loans meet
banks' short-term cash needs

Discount rate > federal funds rate



Let's sum up. In this session, we have discussed the federal reserve's discount window. This is an important part of the traditional monetary policy toolkit. What did we learn? First, discount window loans are short-term secured loans to banks. They help banks manage any short-term cash needs. Second, discount loans are a substitute to borrowing federal funds in the interbank market. As a result, the discount rate provides an upper limit for the federal funds rate.

Lesson 2-3: Non-traditional Monetary Policy

Lesson 2-3.1. Forward Guidance



Learning Objectives

- Forward guidance
- Communication rather than action
- Future path of monetary policy

Hello, and welcome to this lecture on forward guidance. In this class, we will examine a new monetary policy tool, Forward Guidance. Forward Guidance allows the Federal Reserve to communicate future monetary policy intentions. This is communication about tomorrow's action rather than actions taken today. This communication can be about future inflation or economic growth. It often includes how the Fed plans on adjusting interest rates in the future. The goal is to influence market expectations and reduce uncertainty. We will discuss several examples of forward guidance. We will understand why the Fed uses forward guidance, how it influences financial conditions. Let's get started.

What is Forward Guidance?

“Unconventional” tool since 2009

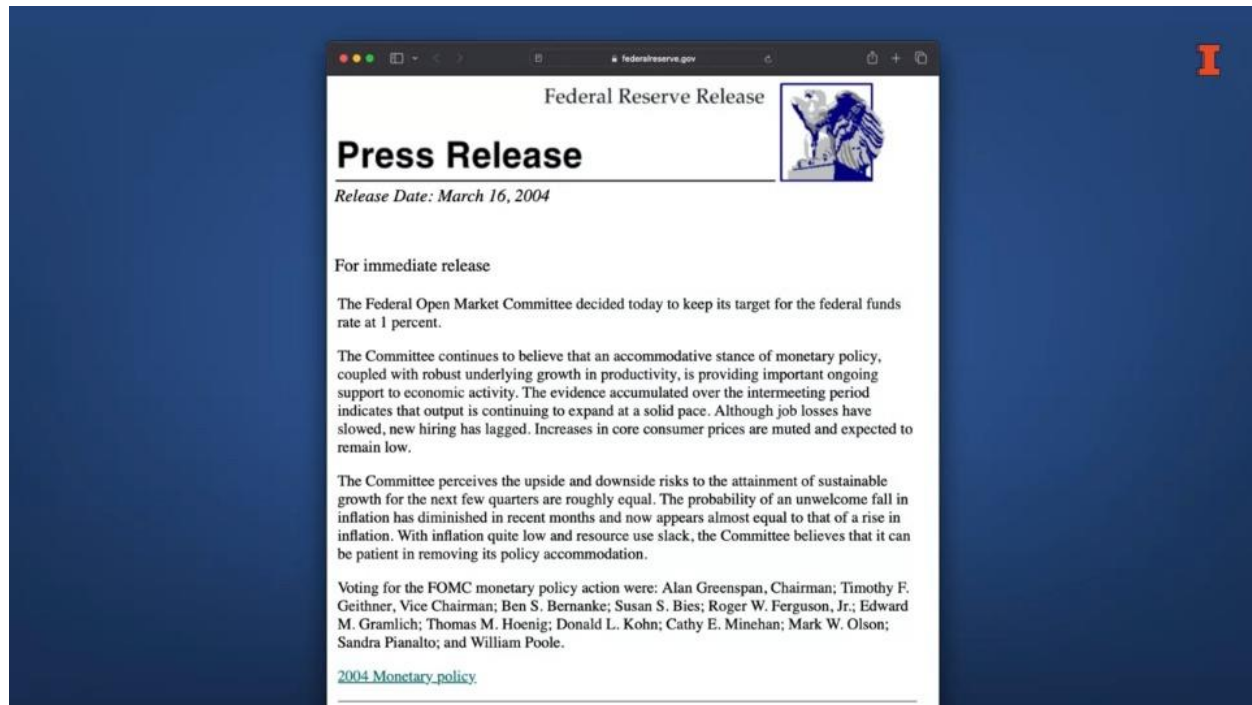
Official FOMC communication

Press releases, speeches, etc.

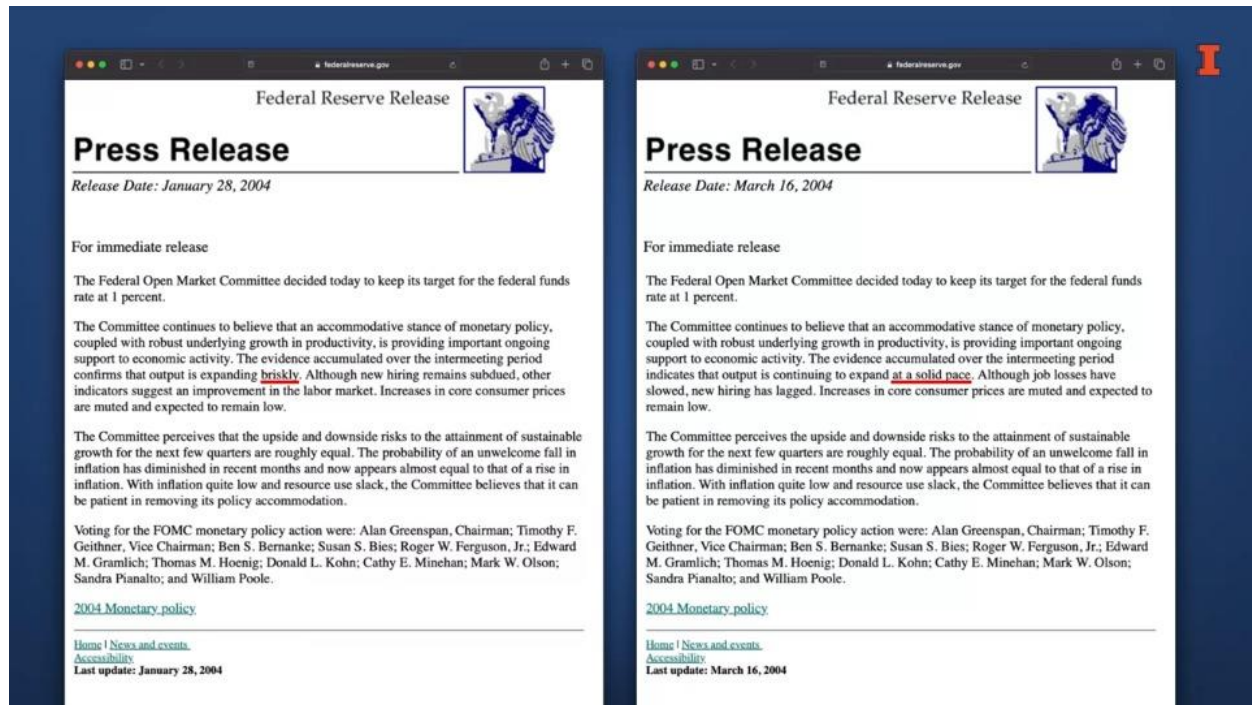
Signals future monetary policy actions



Forward guidance is an unconventional monetary policy tool. Unconventional because it goes beyond the traditional conduct of monetary policy, traditional monetary policy operates on short-term interest rates. In practice, this means open market operations and discount window lending. Forward guidance is relatively new. It arrived in 2009 when the federal funds rate was close to zero. At this point, targeting short term interest rates was no longer an option. To overcome this challenge, the Federal Open Market Committee, or FOMC, decided to communicate how it intended to adjust monetary policy in the future. The Fed tried to target long-term interest rates. Forward Guidance is the official FOMC communication. It may take the form of press releases, speeches, testimony of officials in Congress and so on. These are usually very carefully crafted statements to the public. The goal is to avoid confusion. Forward Guidance signals monetary policy intentions to the public. These can be actions over the coming months or years. These actions can depend on how the economy is evolving. The goal might be to move longer term interest rates. Forward Guidance can also reduce uncertainty.



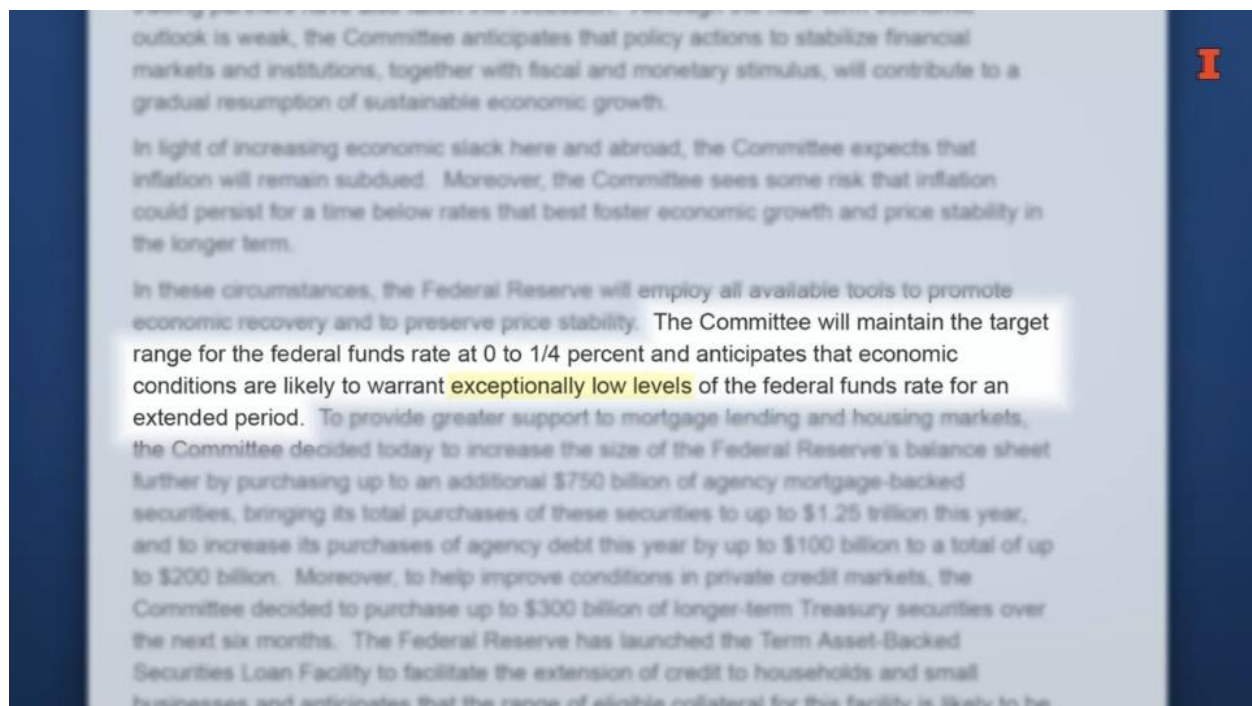
Let's first look at an early example of how the Fed communicates. Since 1994, the FOMC has issued statements announcing decisions from the most recent Monetary policy meeting. Before 2009, these statements were brief. Check out this statement from March 16, 2004. It consists of one sentence about the interest rate, one paragraph summarizing current economic conditions, and one paragraph of potential risks. The statement also indicates which FOMC members voted for an action and which members, if any, descended from it. Finally, notice the absence of any indication about future monetary policy actions.



Now let's play Fat Watcher and compare this statement to the previous statement from January 28, 2004. In the second paragraph, the description of the economy slightly changed. In January, the expansion was risk, but in March, it was only an expansion at a solid pace. This indicates the growth rate, albeit still positive, has decreased. Similarly, the description of the labor market is less favorable. The last paragraph is the same, except for the insertion of that. These changes in language are watched closely by the public. What have they learned from comparing these two statements? Over the span of six weeks, the Federal Reserve has changed its opinion about the state of the economy. While the January statement was quite upbeat, the March statement is less so. What does this mean for monetary policy? Probably that the Federal Reserve will not increase interest rates at the next meeting. This is true. Even so no clear statement about future monetary policy is included.



Now, let's look at a more recent FOMC statement. Here we can see explicit forward guidance. Here's a statement from March 18, 2009. At this point, the federal funds rate was close to zero.



What does the FOMC say? The committee will maintain the target range for the federal funds rate at zero to a quarter percent and anticipates that economic conditions are likely to warrant exceptionally low levels of the federal funds rate for an extended

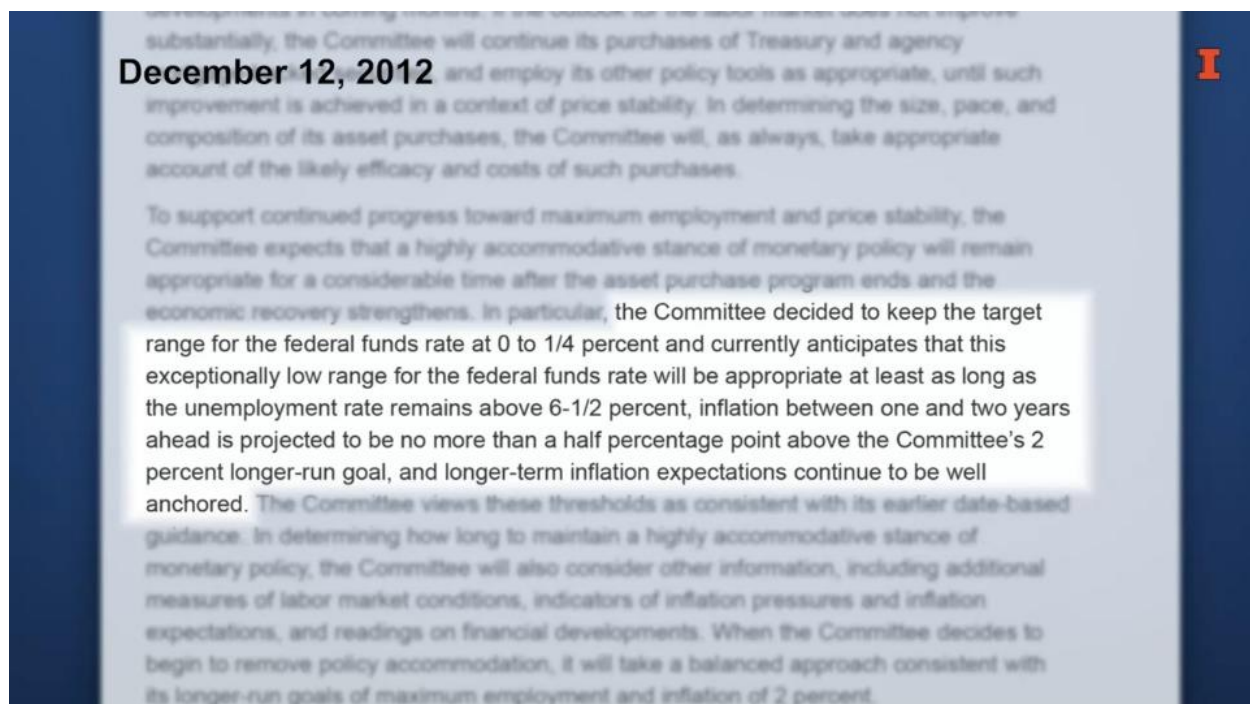
period. This is Forward Guidance. The statement signals that the federal funds rate will be kept at an exceptionally low level, meaning zero for an extended period. The time frame is not exactly clear. How should the public interpret this? Well, financial market analysts would probably expect the federal funds rate to remain at zero for at least one year and possibly much longer.



Now consider this statement from December 12th, 2012. At this point, the federal funds rate was still close to zero, but raising rates was a possibility. You will notice that this statement is much longer. The FMC explains more about the economy.

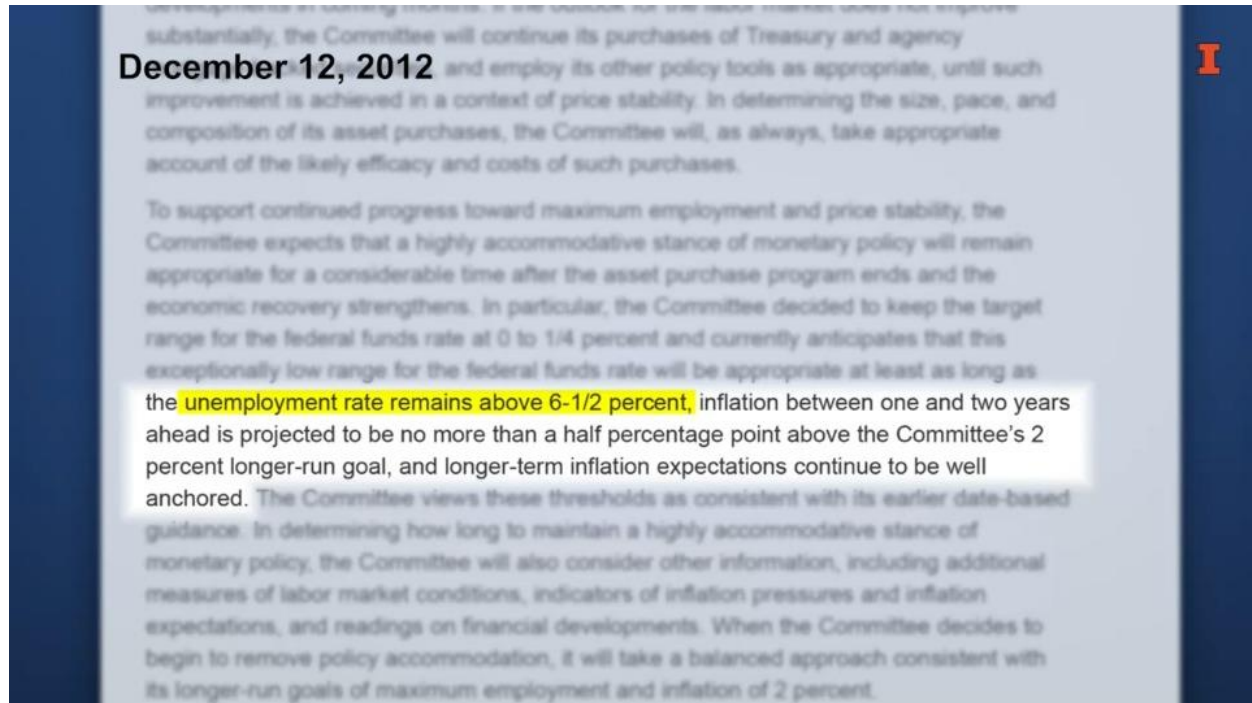


They refer to specific sectors such as households and business investments. But here, the FMC does something new.



In paragraph 5, it states that the committee decided to keep the target range for the federal funds rate at zero to 40%, and currently anticipates that this exceptionally low range for the federal funds rate will be appropriate, at least as long as the unemployment rate remains above 6.5%, inflation between one and years ahead is

projected to be no more than half a percentage points above the committee's 2% longer run goal. And longer-term inflation expectations continue to be well anchored. This is quite a sentence.



Let's unpack this. The FMC says that the target federal funds rate will remain at zero as long as, first, unemployment is above 6.5%. Second, expected inflation in one or two years is not above 2.5% and third, inflation expectations remain well anchored. This means that everyone still expects 2% inflation in the long run. In other words, the FMC would keep the interest rate low under these precise conditions. This type of forward guidance gives planning certainty. It clarifies what extended period means.

Examples of Forward Guidance

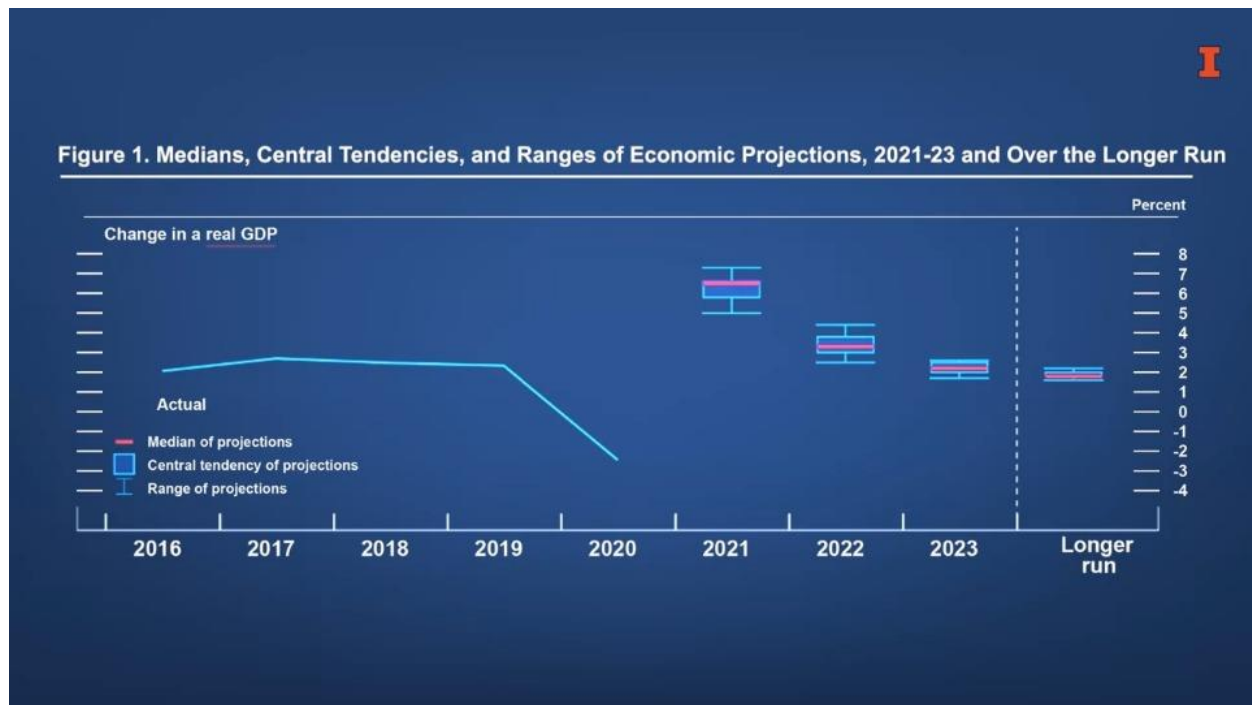
Press statement after FOMC meeting

Minutes from FOMC meeting

Summary of Economic Projections

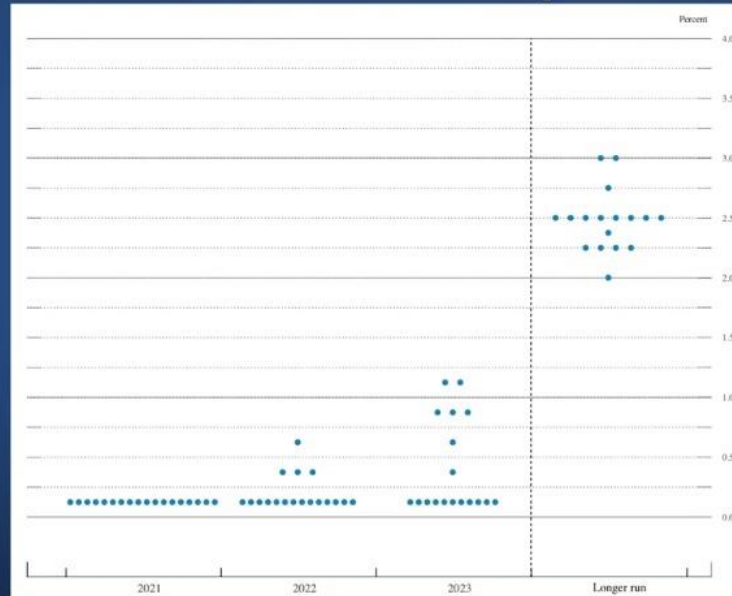


The FMC also releases minutes from its rate setting meetings. These are word for word discussions in the meeting. They arrive three weeks after each meeting. The minutes give further insights about the policy related topics that were discussed, the views expressed by participants, risks and uncertainty about the outlook, and the reasons for the committee's decisions. Of course, financial market analysts come through these minutes to gain insights about future monetary policy. Since 2007, the FMC has also published its summary of economic projections. This comes out four times each year. The projections provide FMC participants assessments of future economic outcomes. This includes GDP, unemployment, inflation, and interest rates. These outcomes are considered over the medium and longer run. Each FMC participant provides an assessment. This allows the public to see a range of forecasts.



Let's have a look at the summary of economic projections from March 17th, 2021. At the time the economy was reeling from the COVID-19 pandemic. Here's the chart for changes in real GDP. This is GDP growth accounting for inflation. You can see that the forecast for 2021 varies between 5% and 7.3%. The difference in the short-term forecast reflects the uncertainty about the recovery. In the long run, the estimates of the participants are really close to each other. There's little disagreement about the long run.

FOMC Interest Rate Expectations



Also, from the March 2021 summary of economic projections, we have the so-called dot plots. This chart summarizes where the FMC thinks the federal funds rate will go. Financial analysts pay close attention to this forward guidance. At that point in time, interest rates were close to zero. The question was, how quickly would rates increase? In 2021, no participant expected an increase from zero. By 2022, four participants think that the economy will have recovered sufficiently to increase the federal funds rate at least once. By 2023, almost half of the participants see the federal funds rate increasing. In the longer run, participants expect the federal funds rate to be 2% and 3%. Of course, these forecasts are subject to revisions. However, they provide valuable insights in the FMC participant thinking. Forward guidance of this kind allows for more planning certainty for financial markets participants.

Summary

Forward guidance is communication

Signals future monetary policy

Allows Fed to:

1. Target long-term interest rates
2. Reduce uncertainty



Let's wrap up. In this lesson, we introduced the concept of forward guidance and discussed several examples. Forward guidance is a monetary policy tool. It allows the Central Bank to signal how it expects the economy and therefore its monetary policy to evolve. The public, especially financial market participants, pay close attention. Routine forward guidance emerged in 2009 when short-term interest rates were very low. At the same time, the Federal Reserve introduced forward guidance to put pressure on long-term interest rates.

Lesson 2-3.2. Large-Scale Asset Purchases

Learning Objectives

Large-scale asset purchases (LSAPs)

Also known as quantitative easing (QE)

Target long-term interest rates



Hello, and welcome to this class on large-scale asset purchases, or LSAPs. LSAPs are often called quantitative easing or QE. They are a form of unconventional monetary policy designed to directly affect longer term interest rates. The Federal Reserve used this tool for the first time during the 2008 financial crisis. At that time, the Fed was trying to stimulate the economy, even so short-term interest rates were at zero. In this session, we will see how LSAPs work. We will look at how these programs were designed. We will see how they affected long-term interest rates and the broader effects for the economy.

Background

Traditional monetary policy $\Rightarrow$ short-term rates

In 2008:

Deep recession

Policy rate near zero

Need new tools



Let's get started. Traditional monetary policy acts only on short-term interest rates, typically the federal funds rate or the discount rate. When the Fed cuts the short-term interest rate, this can also reduce longer term interest rates. However, long-term rates move to a lesser extent. In late 2008, the US economy was entering a deep recession. The Federal Reserve wanted to stimulate the economy. However, the federal funds rate had already been cut to zero. New approaches to monetary policy were needed.

Large-scale asset purchases

Fed buys long-term bonds

QE1, QE2, QE3, Operation Twist

Why?

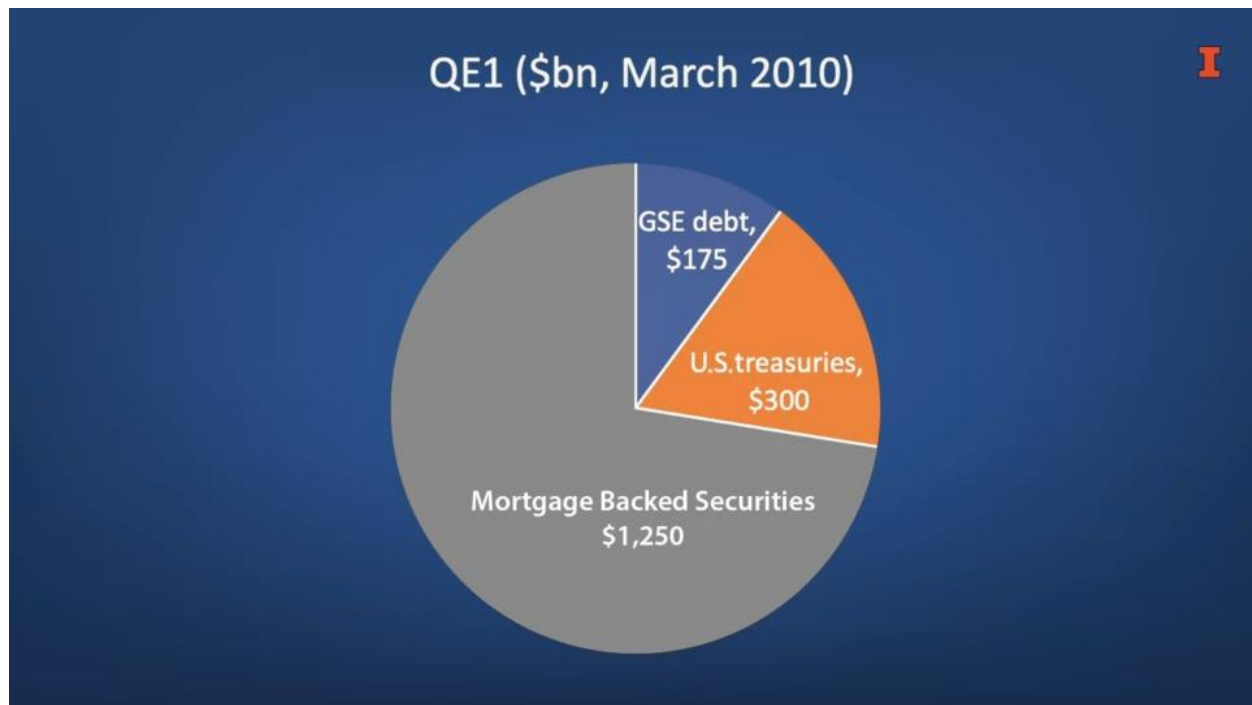
Excess demand
increases bond prices

Higher prices $\Rightarrow$
lower long bond yields

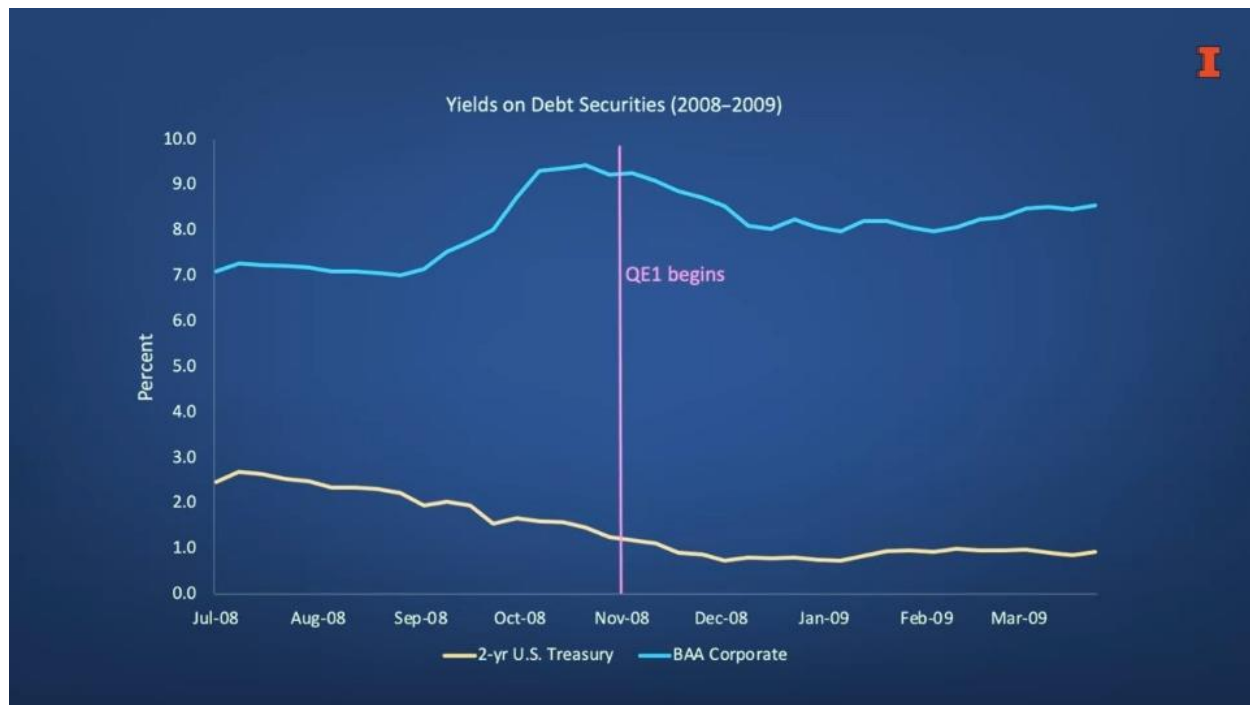
Lower long-term yields $\Rightarrow$
economic stimulus



How did the Central Bank respond to this challenge? The Federal Reserve began purchasing a large amount of longer-term bonds. These were the so-called QE programs. There were four programs altogether QE1, QE2, QE3 and Operation Twist. Why did the Fed do this? How do these purchases affect financial conditions? Well, these purchases acted like a large demand shock for long-term bonds. Excess demand increases prices. Higher prices mean lower bond yields. Remember, there's an inverse relationship between bond prices and bond yields. So, these purchases aim to reduce long-term bond yields directly. Lower long-term bond yields reduce borrowing costs for longer term investment horizons. This should kickstart investment and economic activity.



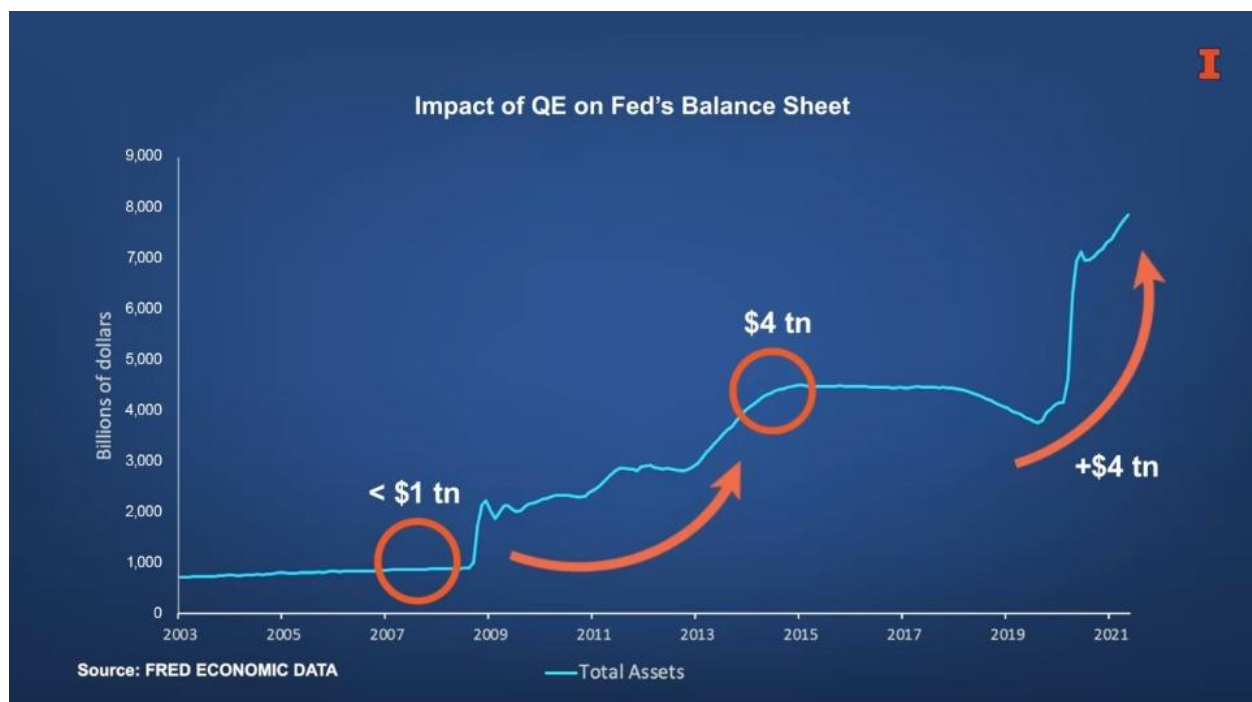
Let's look at these programs in the context of QE1. We can see the scale of the program and understand if it was effective or not. QE1 was launched in November 2008 and expanded in March 2009. By the time it ended in March 2010, the Fed had bought 1.25 trillion in mortgage-backed securities, 175 billion in debt issued by GSEs, Fannie Mae and Freddie Mac and 300 billion in US Treasury securities. These purchases accounted for more than 20% of the market value at the time. This is a massive intervention by a Central Bank. Was the first round of quantitative easing successful? Let's look at long-term yields.



First, long-term Treasuries. The chart shows the yield on 2-year Treasury securities. After the program was implemented, the yield dropped from 1.6% in late October 2008 to 1% in April 2009. Let's look what happened to corporate bonds. These are BAA rated corporate bond yields. This is the borrowing cost for businesses with limited access to capital. As the chart shows, with the rollout of QE1, yields dropped from a peak of 9.5% to about 8%. This lowered the borrowing costs of corporations. We can also check the mortgage market. The Fed made a massive purchase of mortgage-backed securities, or MBS. There's a close relationship between MBS yields and the mortgage rate. When MBS yields fall, then usually the mortgage rate falls as well. You can see in the chart when the QE1 program started, the 30-year fixed mortgage rate started to fall from almost 6.5% in late October 2008 to 4.85% in early April 2009. This has a sizable effect. Lower mortgage rate encouraged the purchases of homes. This intervention may have helped to support the housing market. The other quantitative easing programs had a similar flavor. QE2 was announced in October 2010. QE3 in September 2012. Operation Twist involved the Federal Reserve selling short-term securities and using the proceeds to buy longer term securities. In each case, the intent was to push down longer-term interest rates. This reduces the cost of borrowing for households and firms. These programs stopped in October 2014. By this time, the labor market had recovered.



The chart shows the impact of the QE programs on the Federal Reserve's balance sheet. Remember, these large-scale asset purchases ended up with the Fed.



By 2014, the Federal Reserve held almost 4 trillion in securities. These were mostly, US Treasuries and MBS. This was a dramatic expansion in the balance sheet relative to the period before the 2008 financial crisis. Large scale asset purchases were also part of the Federal Reserve's response to the COVID-19 pandemic. In March 2020, the FOMC

announced that the Federal Reserve will continue to purchase Treasury securities and agency mortgage-backed securities, in the amount needed to support smooth market functioning and effective transmission of monetary policy to broader financial conditions. This program was another open-ended commitment to support the economy.

Summary

Policy rate at zero $\Rightarrow$ Fed resorts to QE

QE targets long-term interest rates directly

QE reduces long-term borrowing costs
for households and businesses



What have we learned in this class? First, when short-term interest rates are at 0%, the Federal Reserve can use large scale asset purchases, or QE, to conduct monetary policy. Second, large scale asset purchases can target longer term interest rates. Third, large scale asset purchases reduce the costs of credit for consumers and corporations by lowering long-term interest rates.

Lesson 2-3.3. Monetary Policy With Abundant Bank Reserves

Learning Objectives

Post-2008, too many bank reserves

Traditional OMOs ineffective

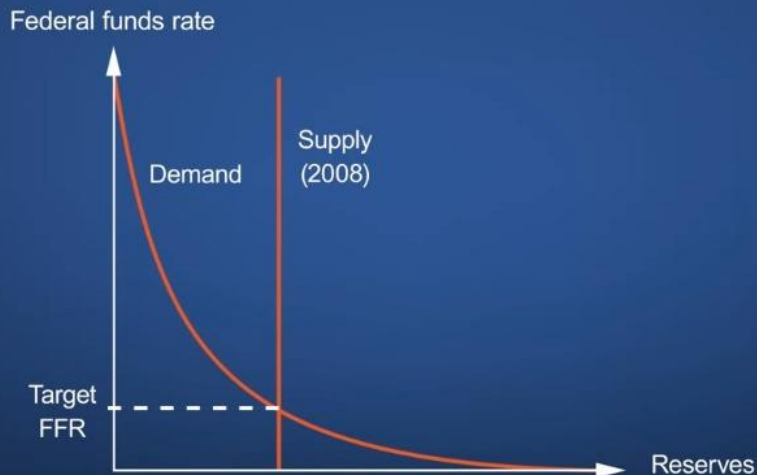
New monetary policy tools:

1. Interest on reserve balances (IORB)
2. Overnight reverse repurchase agreement facility (ON RRP)



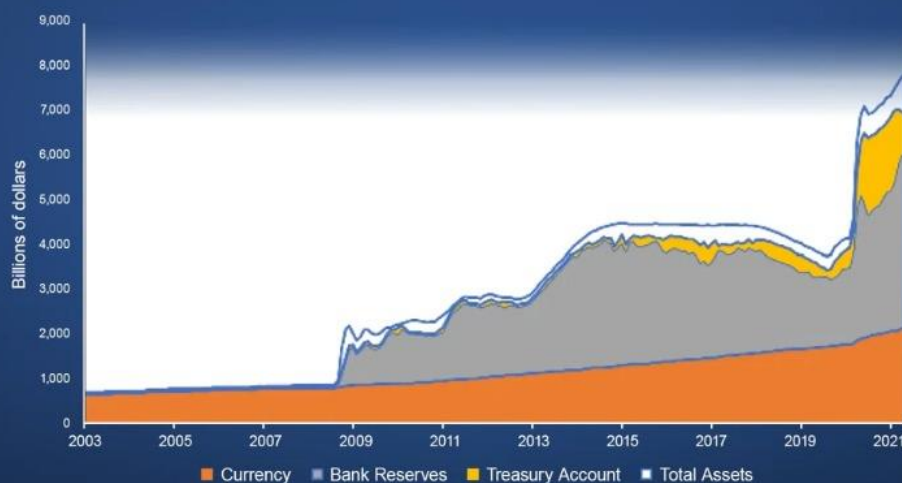
Hello, and welcome to this class on monetary policy implementation when bank reserves are abundant. After 2008, and the large-scale asset purchase programs, the banking system was awash with reserves. This rendered traditional open market operations less effective for managing interest rates. In this class, we examine two new monetary policy tools created for this environment. First, interest on reserve balances, or IORB, and second, the Overnight Reverse Repurchase Agreement facility, or ON RRP. These tools help the Federal Reserve control short-term interest rates when there are high levels of bank reserves. Let's get started.

Federal Funds Market



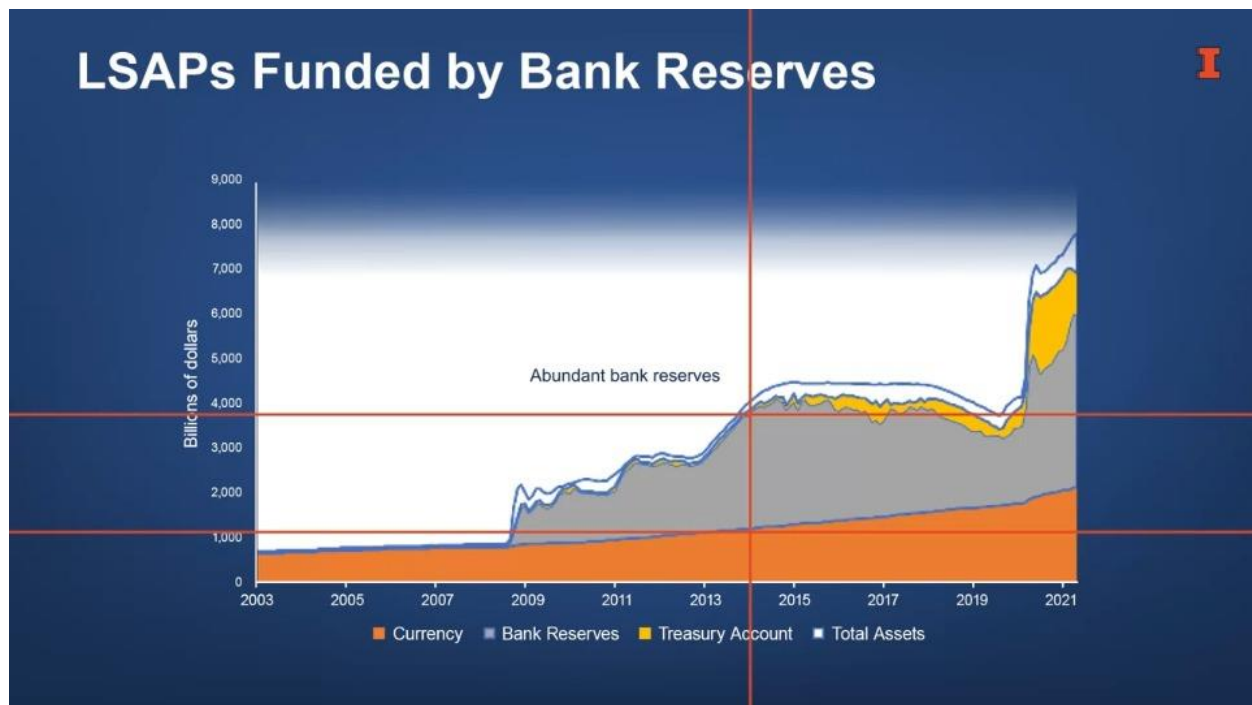
How does the Federal Reserve target interest rates? Historically, the Federal Reserve would conduct open market operations. This involves buying and selling securities to change the level of supply of reserves in the federal funds market. The figure shows the federal funds market. By adjusting the supply of bank reserves, the Federal Reserve controls the federal funds rate.

LSAPs Funded by Bank Reserves

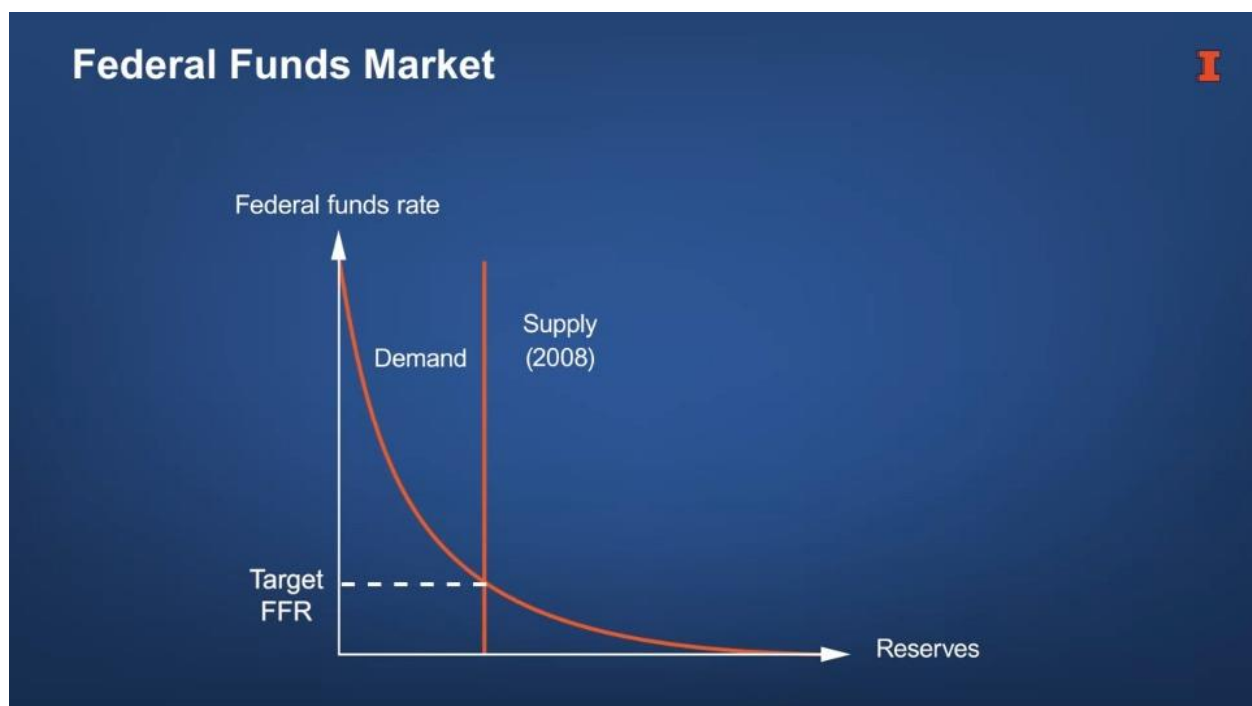


Then the 2008 financial crisis arrived. The federal funds rate was cut to zero, there were

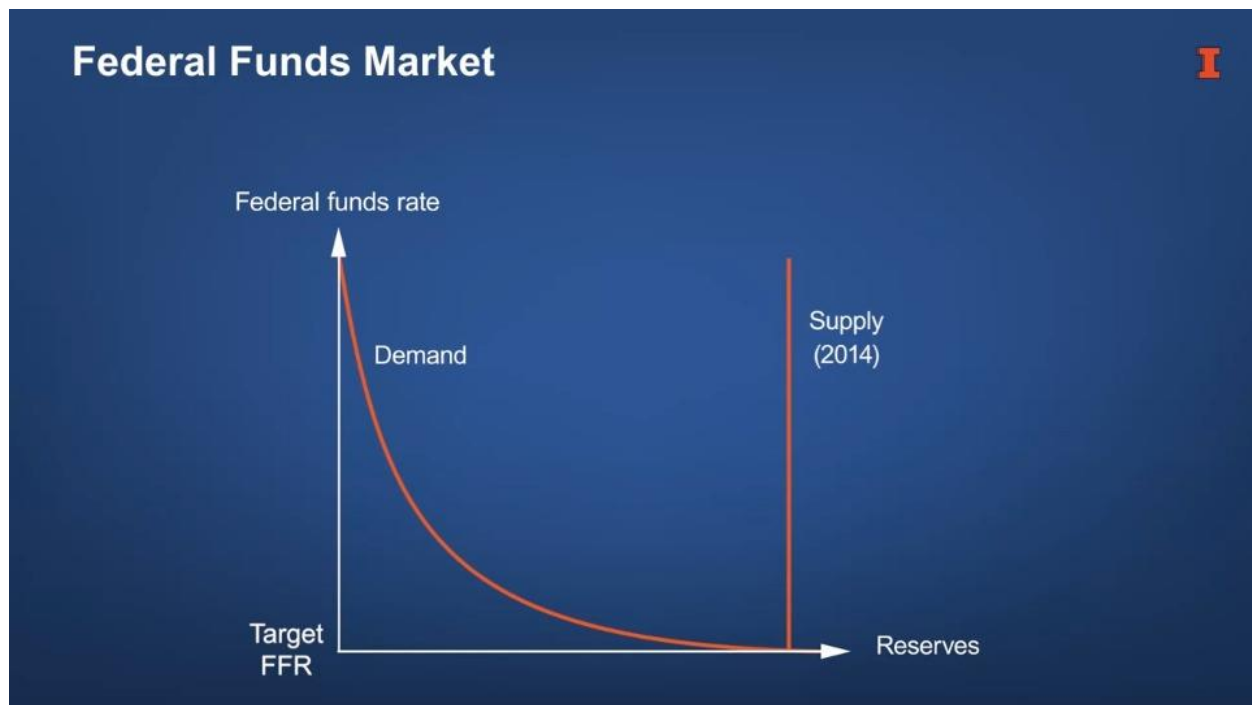
large-scale asset purchases. These purchases were funded by a drastic increase in bank reserves. Reserves had grown from 14 billion in 2007 to over 2.6 trillion in 2014.



We have examined this chart in another lecture.



We can see the impact on the equilibrium in the federal funds market. The supply of reserves is much larger, and the federal funds rate is zero.



Why is this a problem? Eventually, the Federal Reserve would like to increase interest rates. However, the federal funds rate would no longer respond to small changes in bank reserves. Traditional open market operations would not be enough. Large changes in the supply of reserves were needed for the federal funds rate to respond. In theory, the Federal Reserve could reduce reserves through large-scale asset sales. Asset sales, not purchases. However, the fear was that this would be too costly. Unloading \$1 trillion of securities could have a large unintended consequence for financial markets and borrowers. It was deemed too risky.

Monetary Policy With Abundant Reserves

Two tools to control interest rate:

1. Interest on reserve balances (IORB)
2. Overnight reverse repo facility (ON RRP)



Two alternatives were adopted instead. First, the Federal Reserve pays interest on reserve balances, or IORB. This tool targets banks. Second, the Federal Reserve created the overnight reverse repo facility, or ON RRP. The ON RRP targets non-banks, such as money market mutual funds. These non-banks are also active participants in cash markets, so they cannot be ignored. Together, these tools allow the Federal Reserve to control short-term interest rates, even when reserves are abundant.

Interest on Reserve Balances (IORB)

Banks only

Earn IORB for investing overnight with Fed

Fed sets IORB = target FFR, then:

In theory, $FFR = IORB$

In practice, IORB provides a floor



Let's see how this works. We will start with interest on reserve balances. The Federal Reserve now pays interest on reserve balances. These are funds that banks hold overnight in their account at the Fed. These funds are not lent out to other banks in the interbank market. How can this help with monetary policy implementation? Assume the Federal Reserve sets the interest on reserve balances equal to the target federal funds rate. Now, banks are indifferent between investing overnight with the Fed or making an interbank loan. We will earn the target federal funds rate, either way. This must be the case in the equilibrium. If the federal funds rate was higher than the IORB, then banks would lend out reserves in the interbank market. This arbitrage process should continue until the two interest rates equalize. In practice, the interest of reserves provides a lower bound on the federal funds rate.

Federal Funds Market



We can see this graphically. Here's the supply and demand in the federal funds market. By paying interest on reserve balances, this will put a floor on the federal funds rate. In equilibrium, no bank would be willing to accept less than the IORB by lending in the interbank market.

Federal Funds Market



This allows the Federal Reserve to increase the federal funds rate without changing the supply of reserves.

Overnight Reverse Repo Facility (ON RRP)

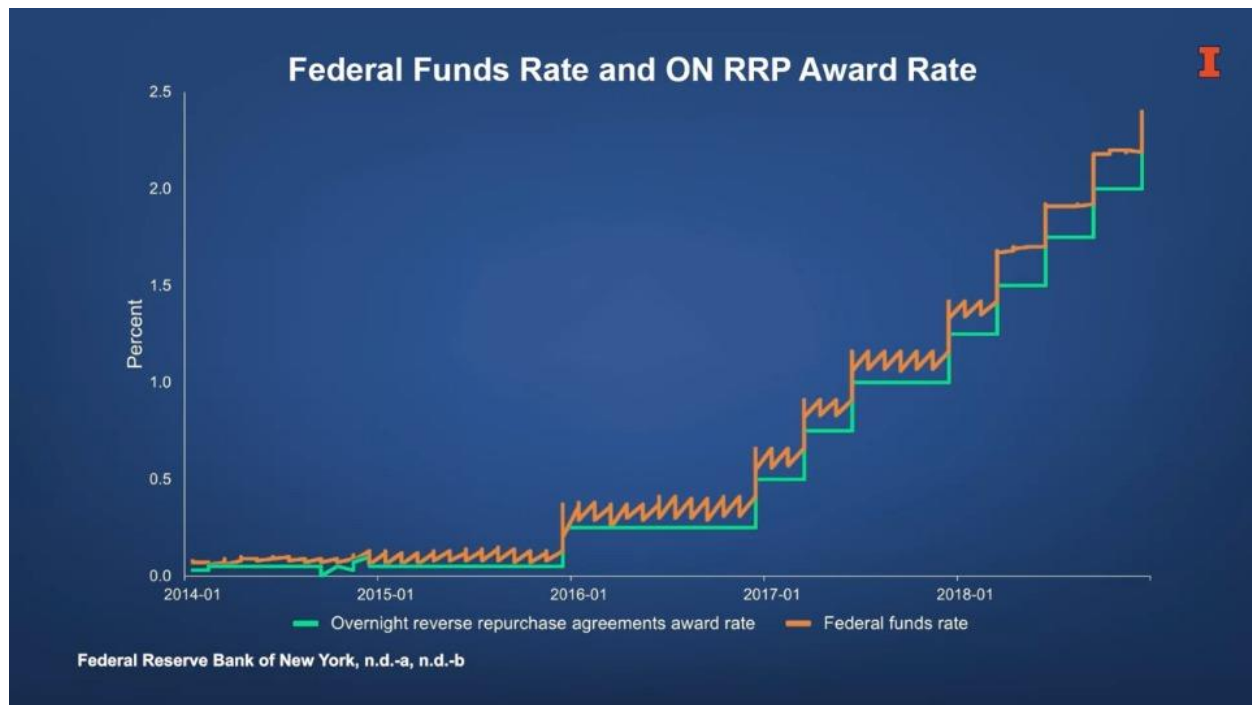
Non-banks as well

Earn ON RRP rate for
investing overnight with Fed

Also establishes floor on FFR



What about the ON RRP facility? Why do we need it? How does it work? In the United States, non-banks play an important role in the cash markets. Money market funds, for example. Since these lenders cannot access the IORB, they might sometimes be willing to lend at a lower rate. This is better than accepting nothing on cash that is sitting idle. This means, short-term interest rates could dip below the IORB. To combat this effect, the Federal Reserve created the ON RRP facility. It is the equivalent of IORB for these non-banks. They can invest overnight with the Fed through a repurchase agreement. This facility ensures that the Fed can set a floor on the short-term interest rates. These non-banks are unwilling to lend for less than the ON RRP rate. Let's see this in action.



Here's the chart showing the ON RRP rate along with the federal funds rate. As you can see, both rates track each other well. This means that the ON RRP can help the Federal Reserve to increase the federal funds rate. Remember, this was a time when there were massive amounts of bank reserves in the system. Under these conditions, a traditional open market operation would not have worked.

Summary

Post-2008, massive quantity of bank reserves

Traditional monetary policy ineffective

Solution: IORB and ON RRP



Let's wrap up. This was a class on how central banks can implement monetary policy

when the system is flooded with bank reserves. We covered two key ideas. First, when there are a lot of reserves, traditional monetary policy tools may fail to influence interest rates. Second, the Federal Reserve created new tools, IORB and ON RRP, to provide a lower bound for the federal funds rate.